



US 20250272473A1

(19) **United States**

(12) **Patent Application Publication**
Roybal et al.

(10) **Pub. No.: US 2025/0272473 A1**

(43) **Pub. Date: Aug. 28, 2025**

(54) **MODEL CREATION AND MANAGEMENT USING A MODEL MANAGEMENT SERVICE**

(52) **U.S. Cl.**
CPC **G06F 40/117** (2020.01); **G06F 40/197** (2020.01)

(71) Applicant: **AT&T Intellectual Property I, L.P.**,
Atlanta, GA (US)

(72) Inventors: **Michael Roybal**, Woodinville, WA (US); **Matthew Heyer**, Saint George, UT (US); **Wilson Huang**, Pleasanton, CA (US)

(73) Assignee: **AT&T Intellectual Property I, L.P.**,
Atlanta, GA (US)

(21) Appl. No.: **18/588,979**

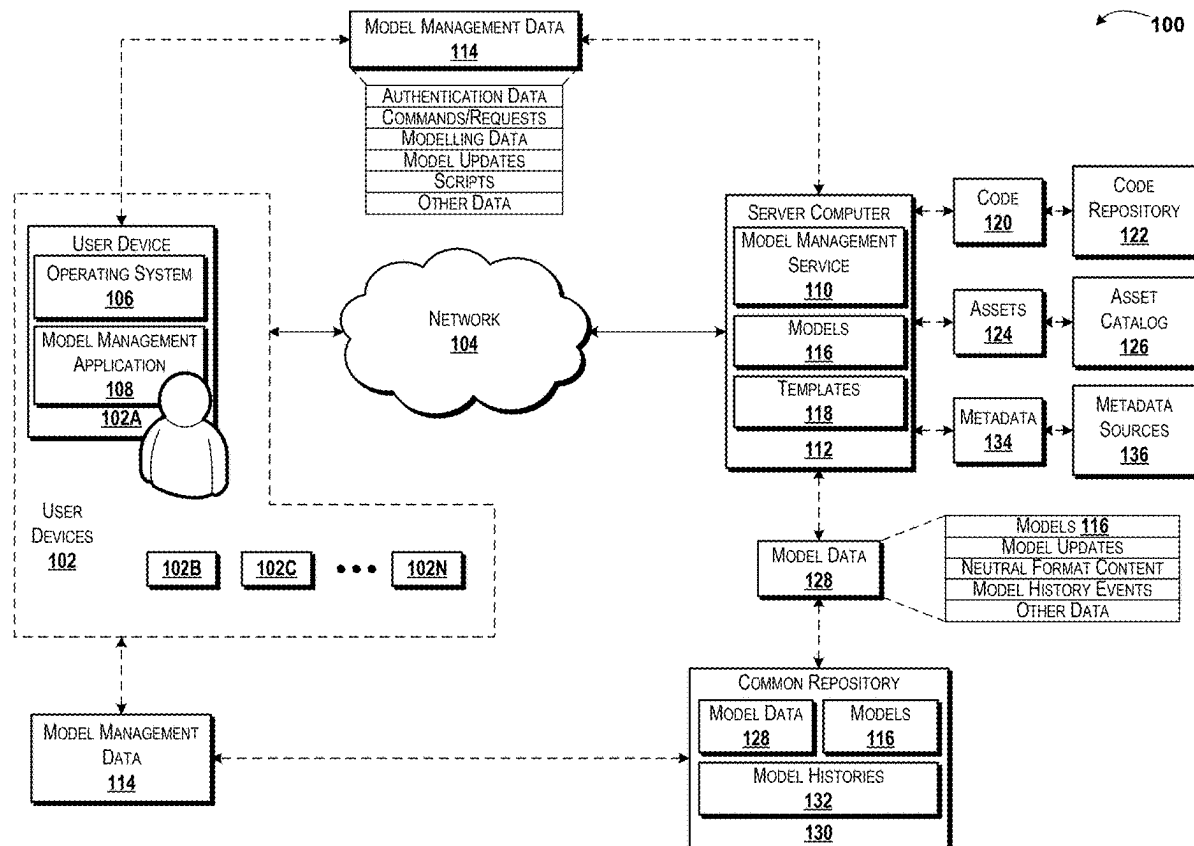
(22) Filed: **Feb. 27, 2024**

Publication Classification

(51) **Int. Cl.**
G06F 40/117 (2020.01)
G06F 40/197 (2020.01)

(57) **ABSTRACT**

Model creation and management using a model management service can include obtaining model management data that comprises modeling data that defines a project to model, the project including a network flow, a network system, a device, or a process; detecting, in the model management data, a selection of a template to apply to the project to generate the model; and creating, based on the model management data and the template, the model. The model can include a directory, a data structure, and a diagram that describes the project. The model can be converted into a device-agnostic format version of the model and loaded to a data storage resource as model data that can include the device-agnostic format version of the model.



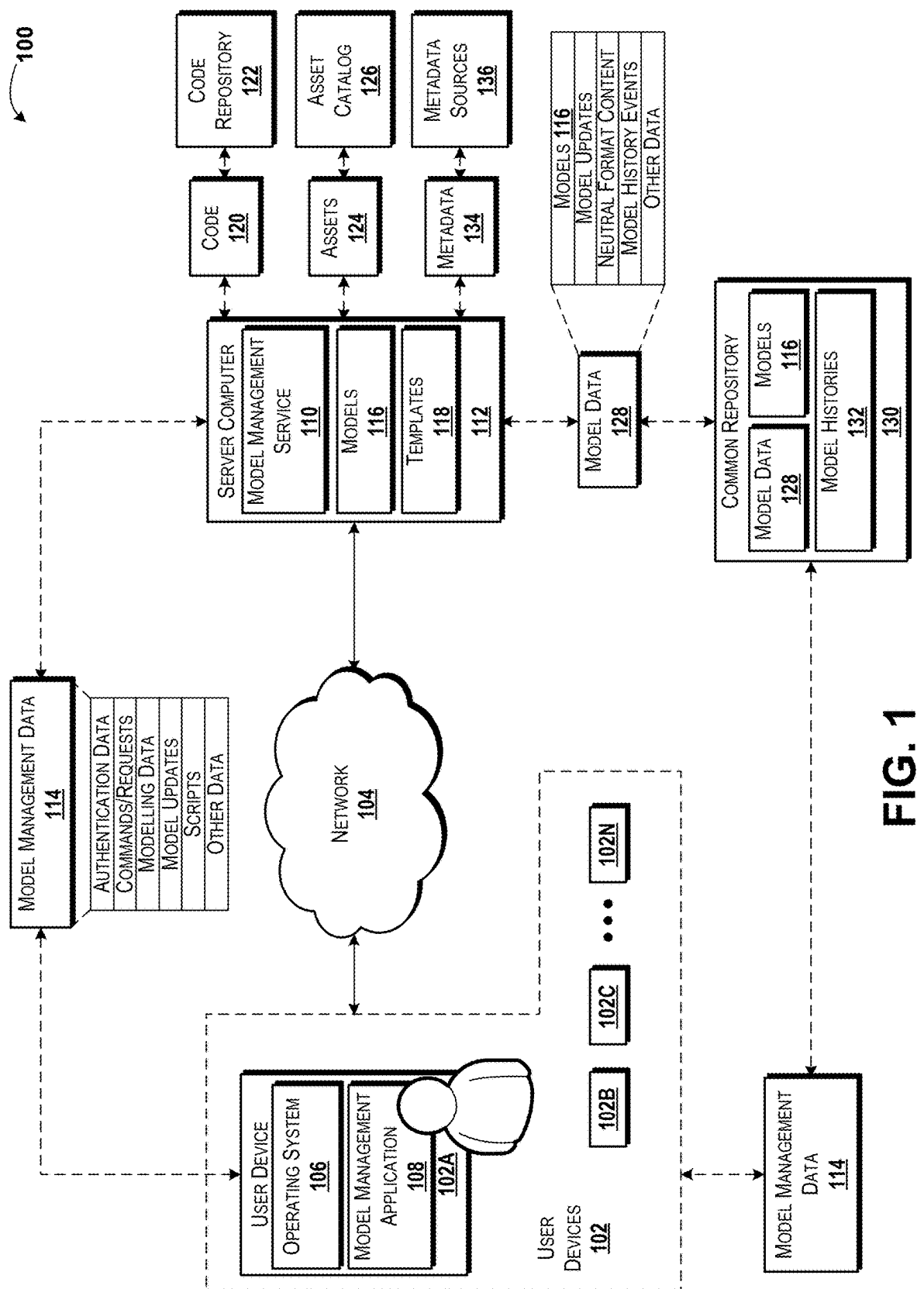
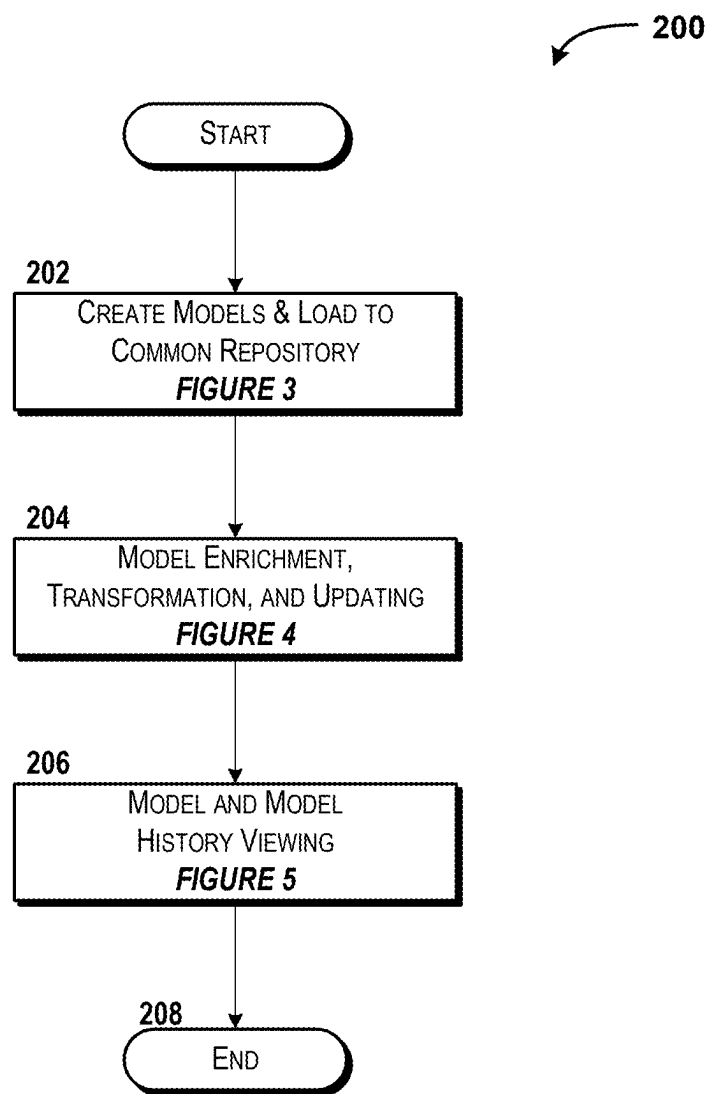


FIG. 1

**FIG. 2**

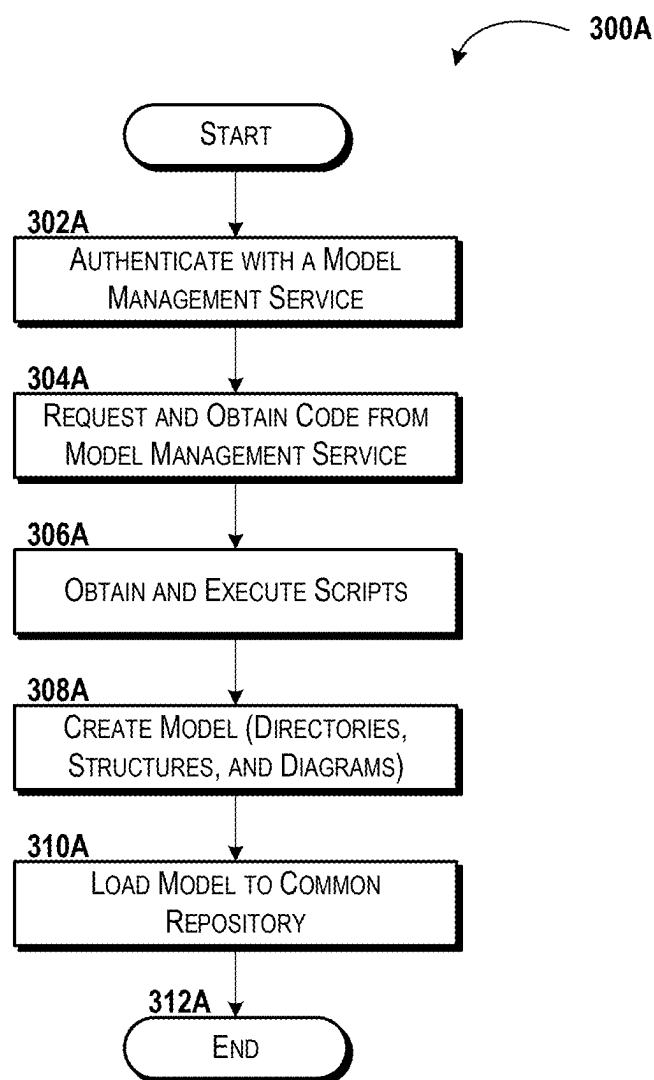


FIG. 3A

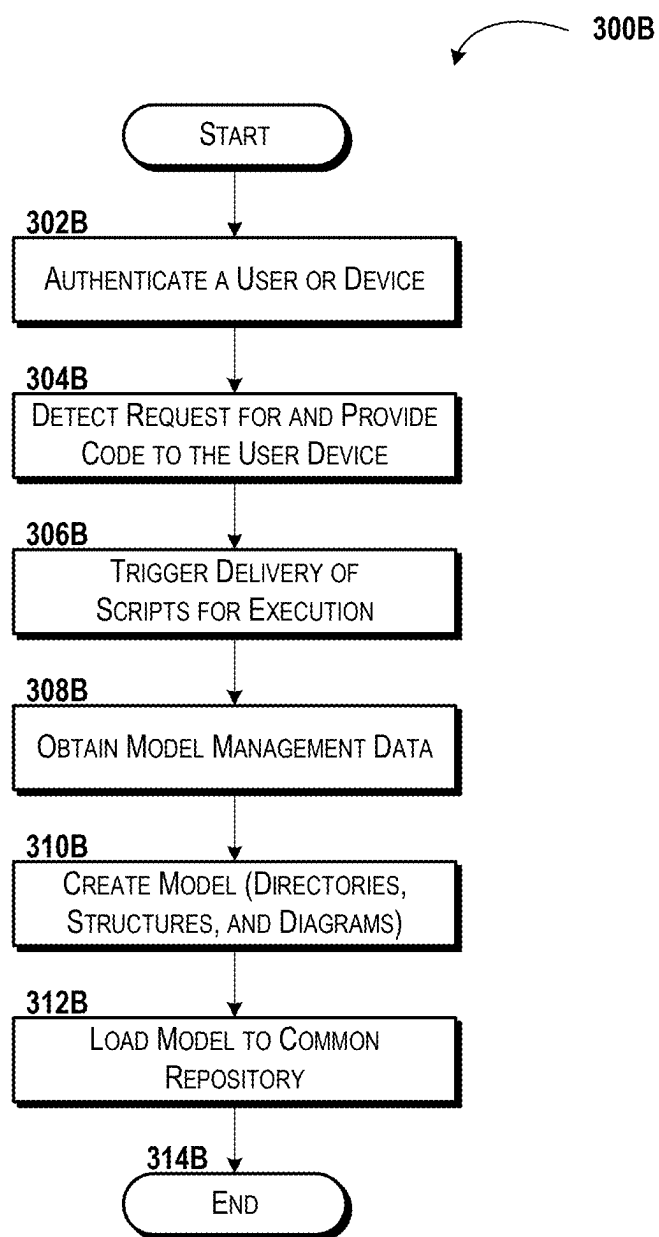


FIG. 3B

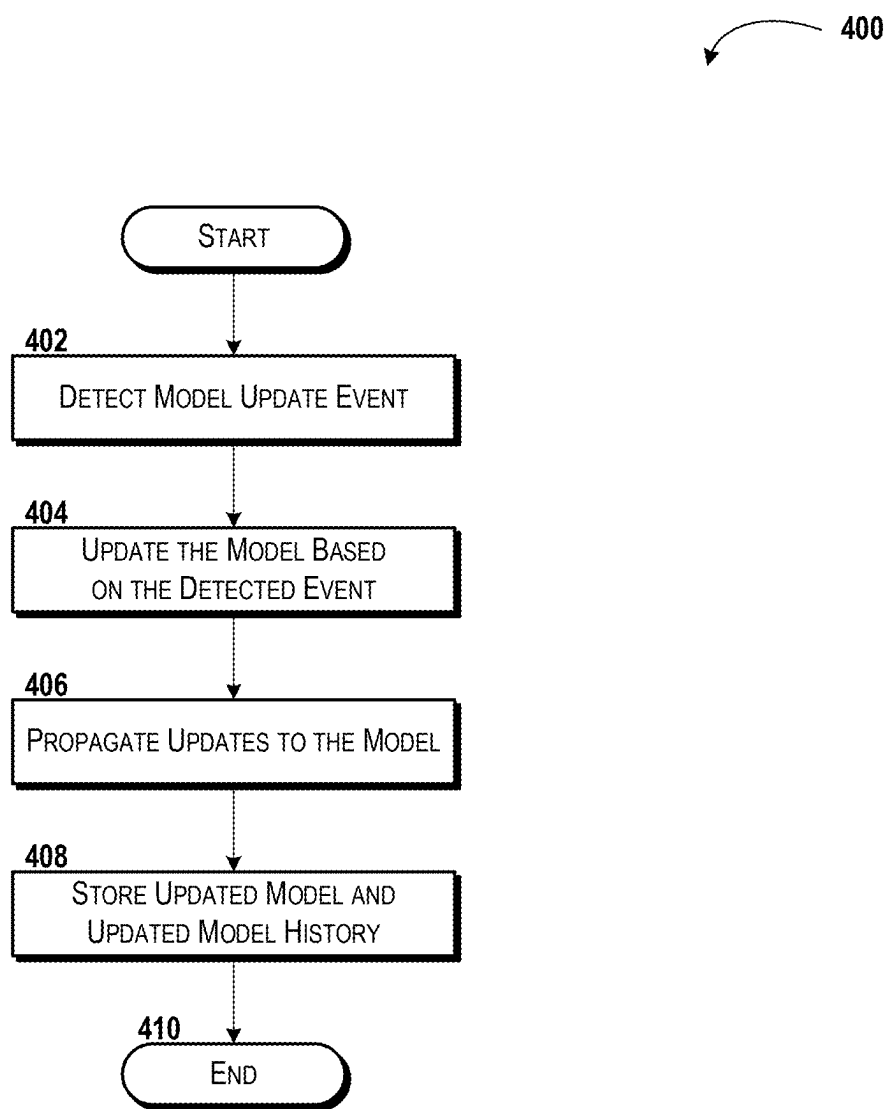


FIG. 4

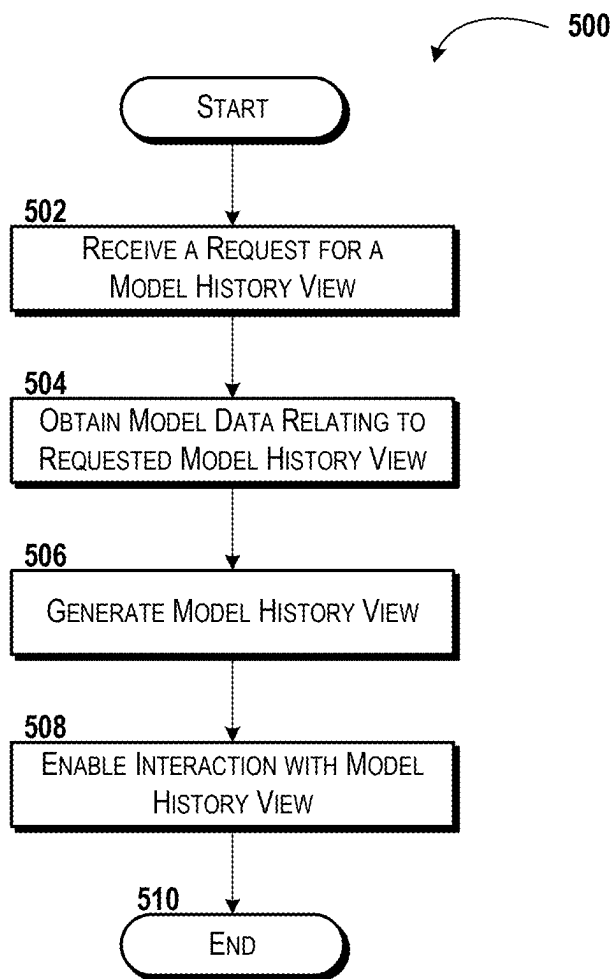


FIG. 5

600A

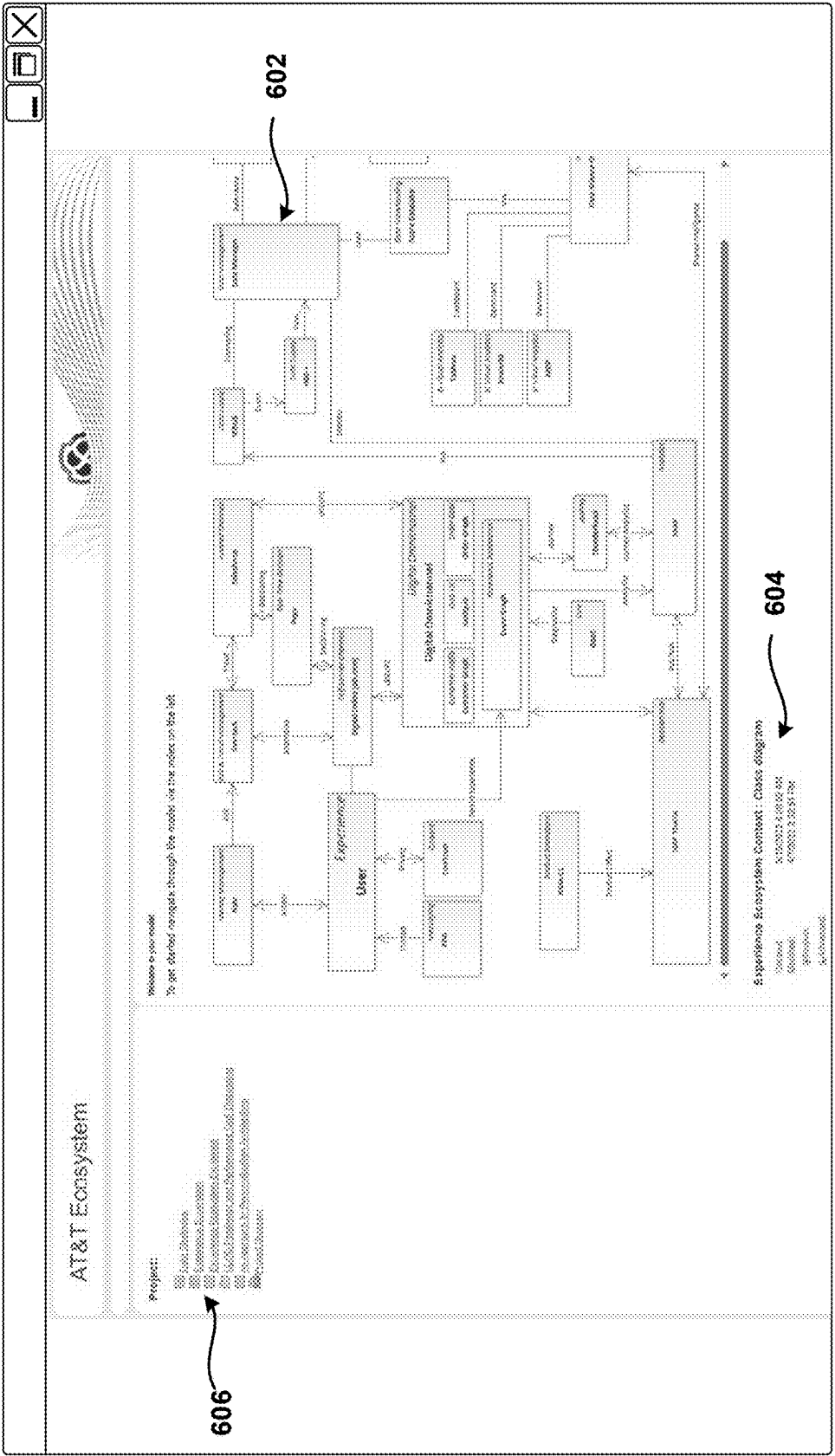


FIG. 6A

600B

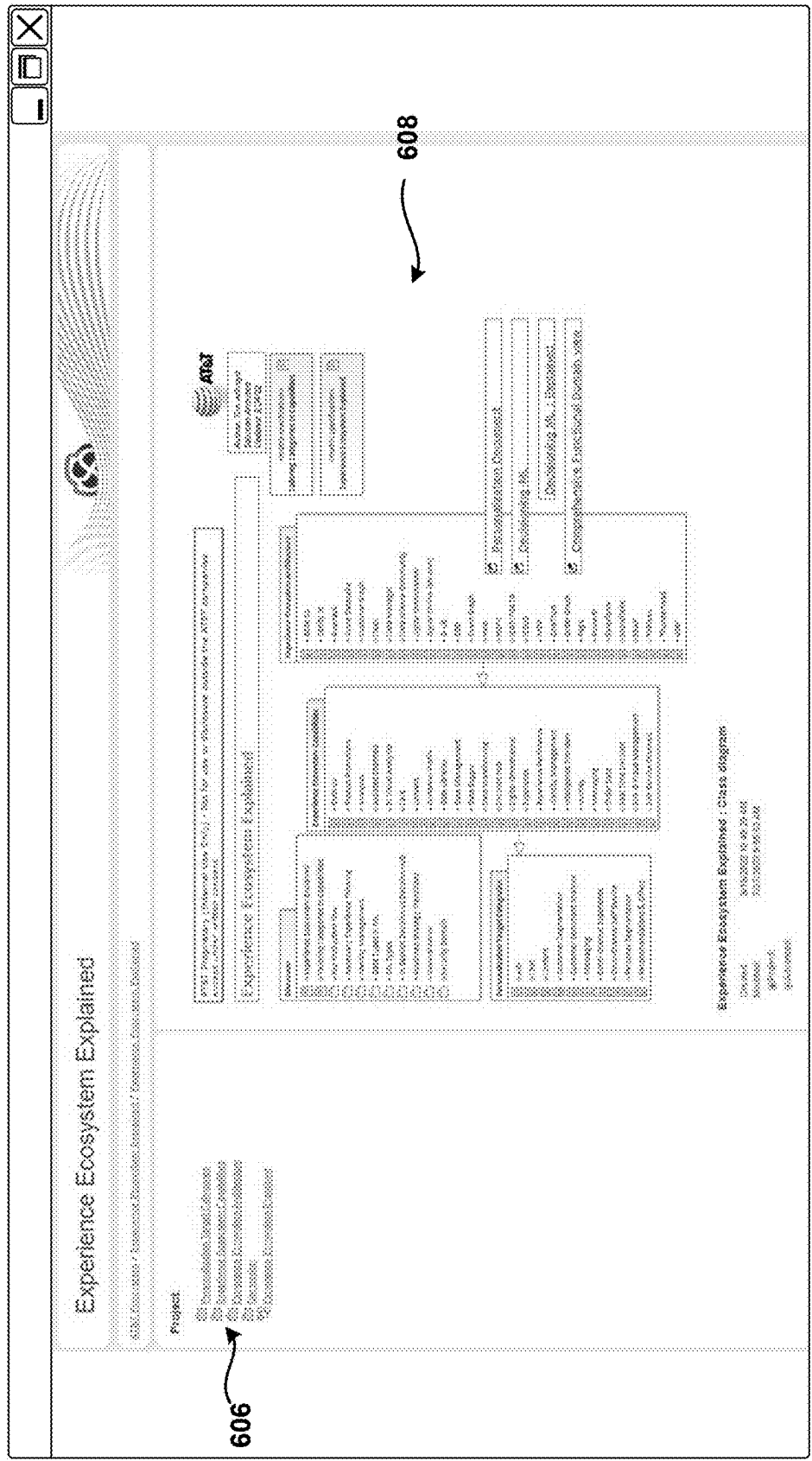


FIG. 6B

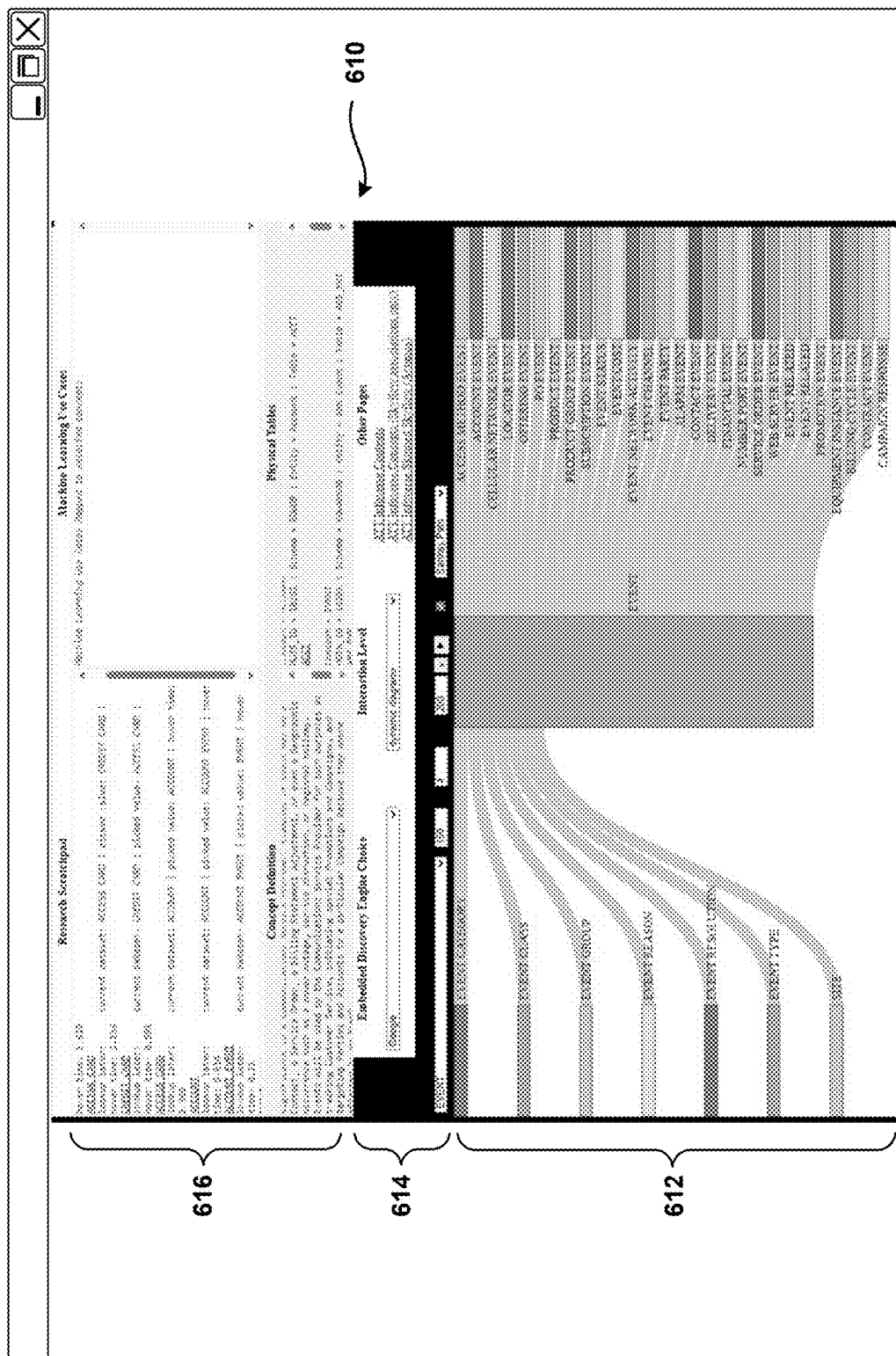


FIG. 6C

700A

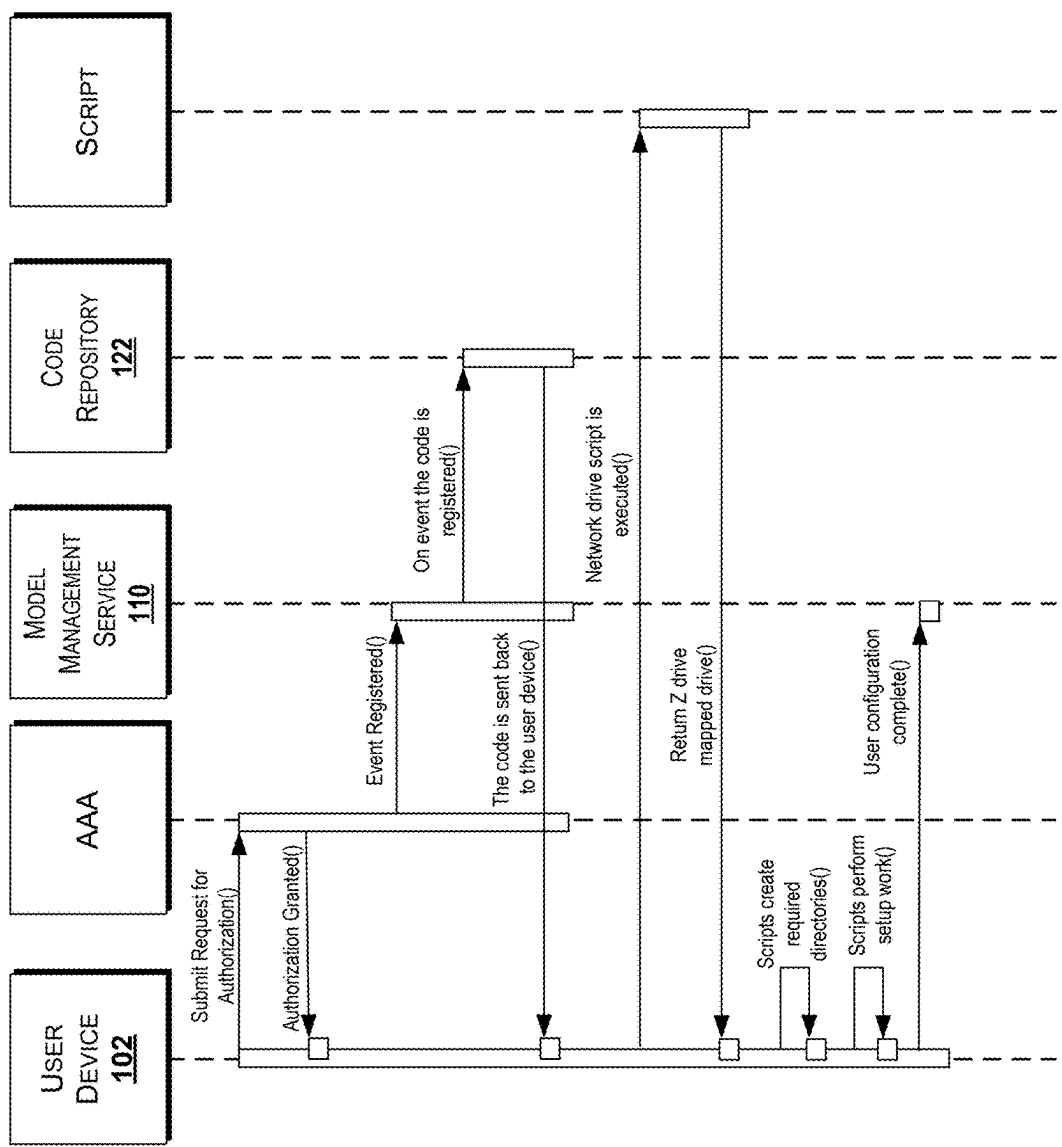


FIG. 7A

700B

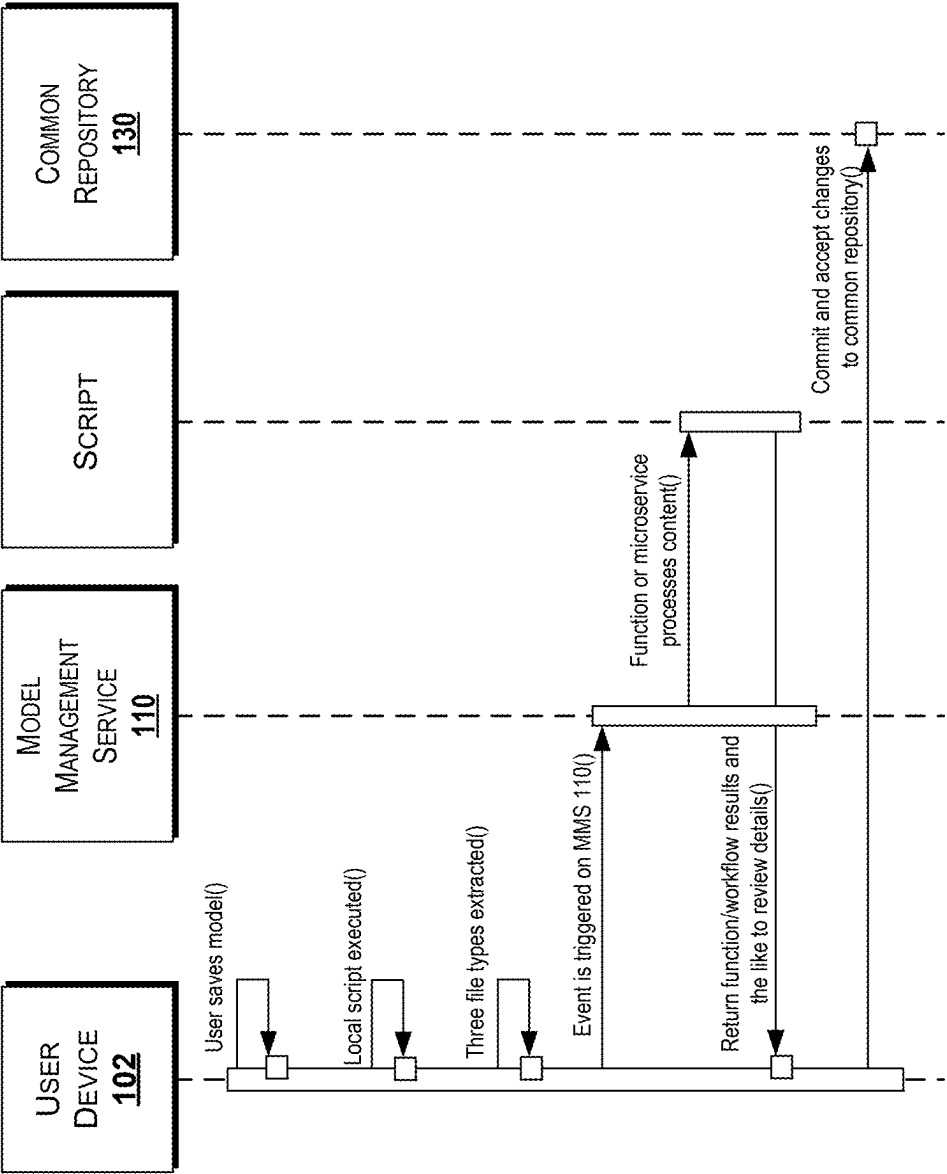


FIG. 7B

700C

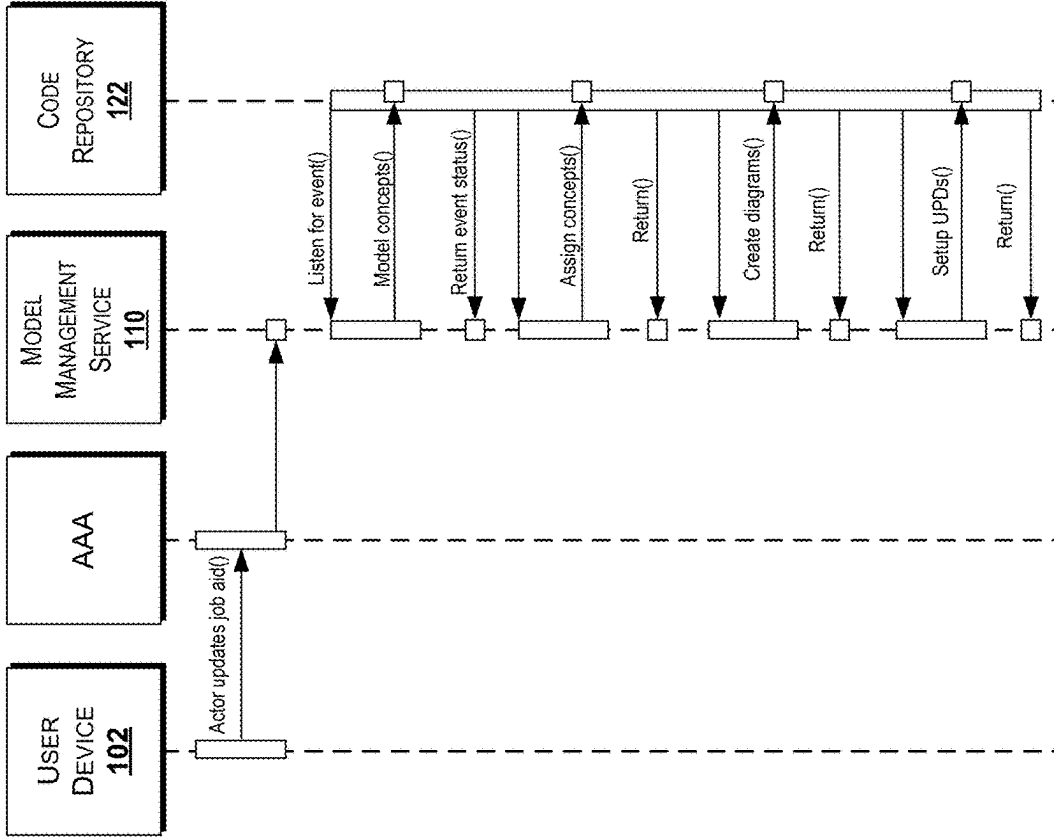


FIG. 7C

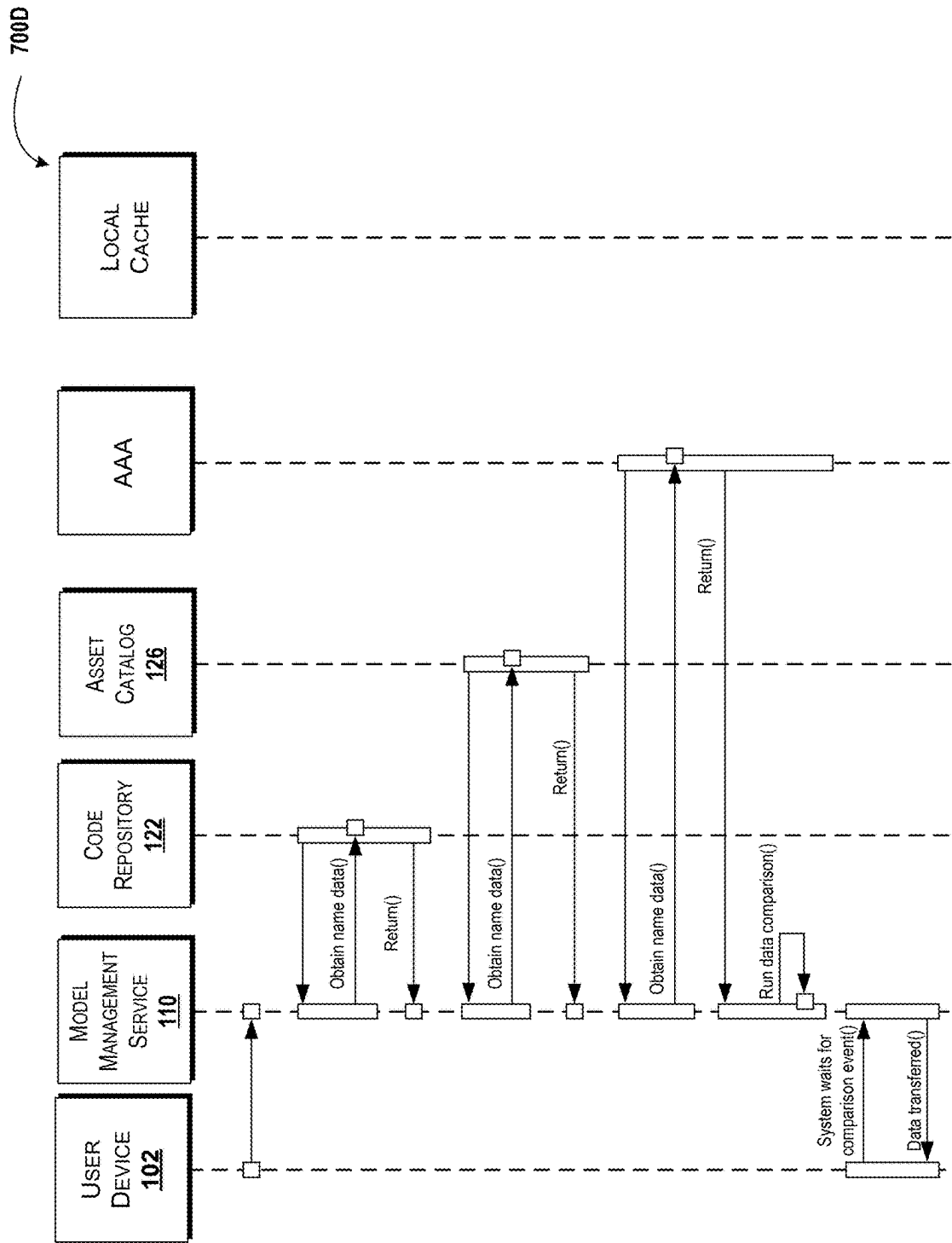


FIG. 7D

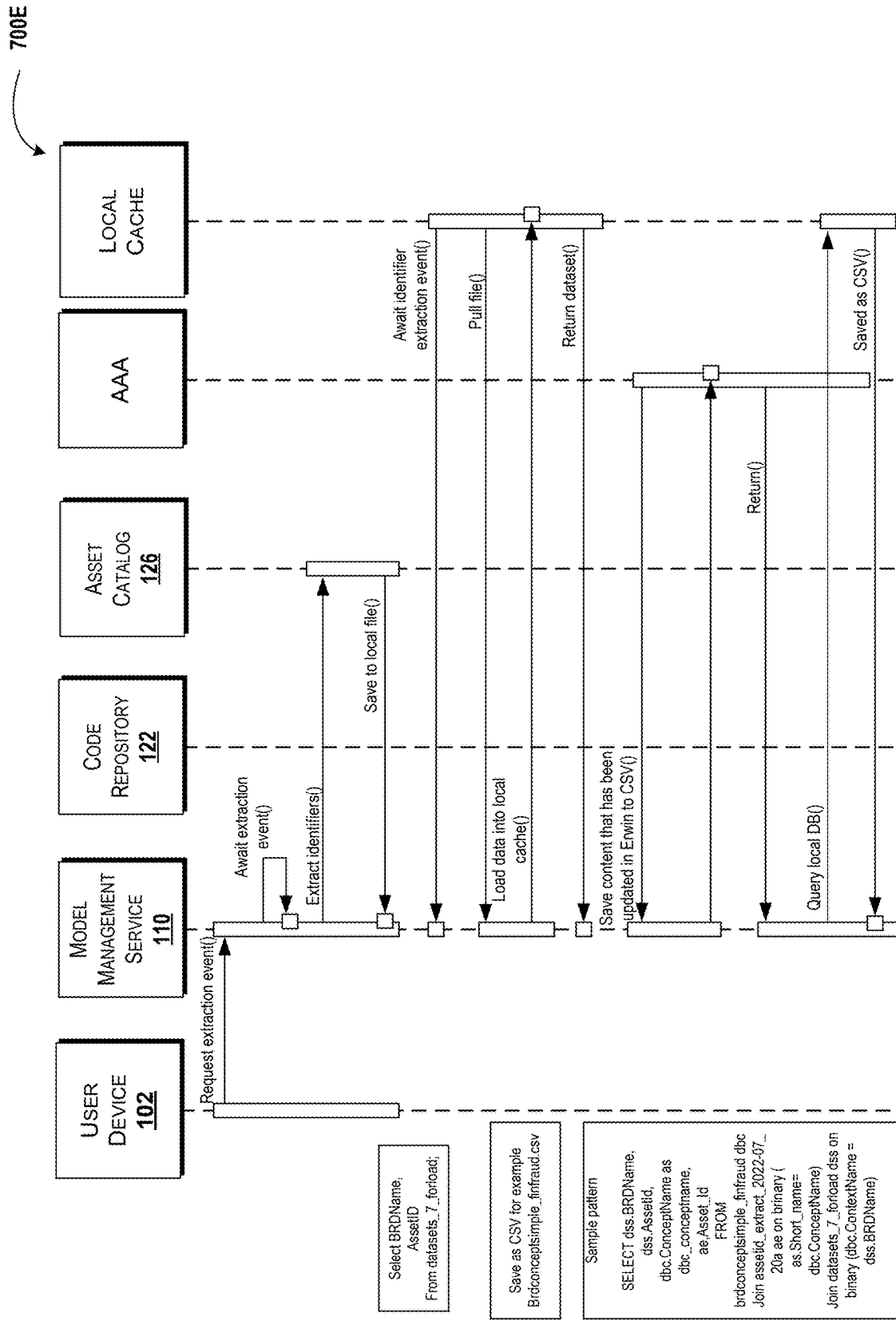


FIG. 7E

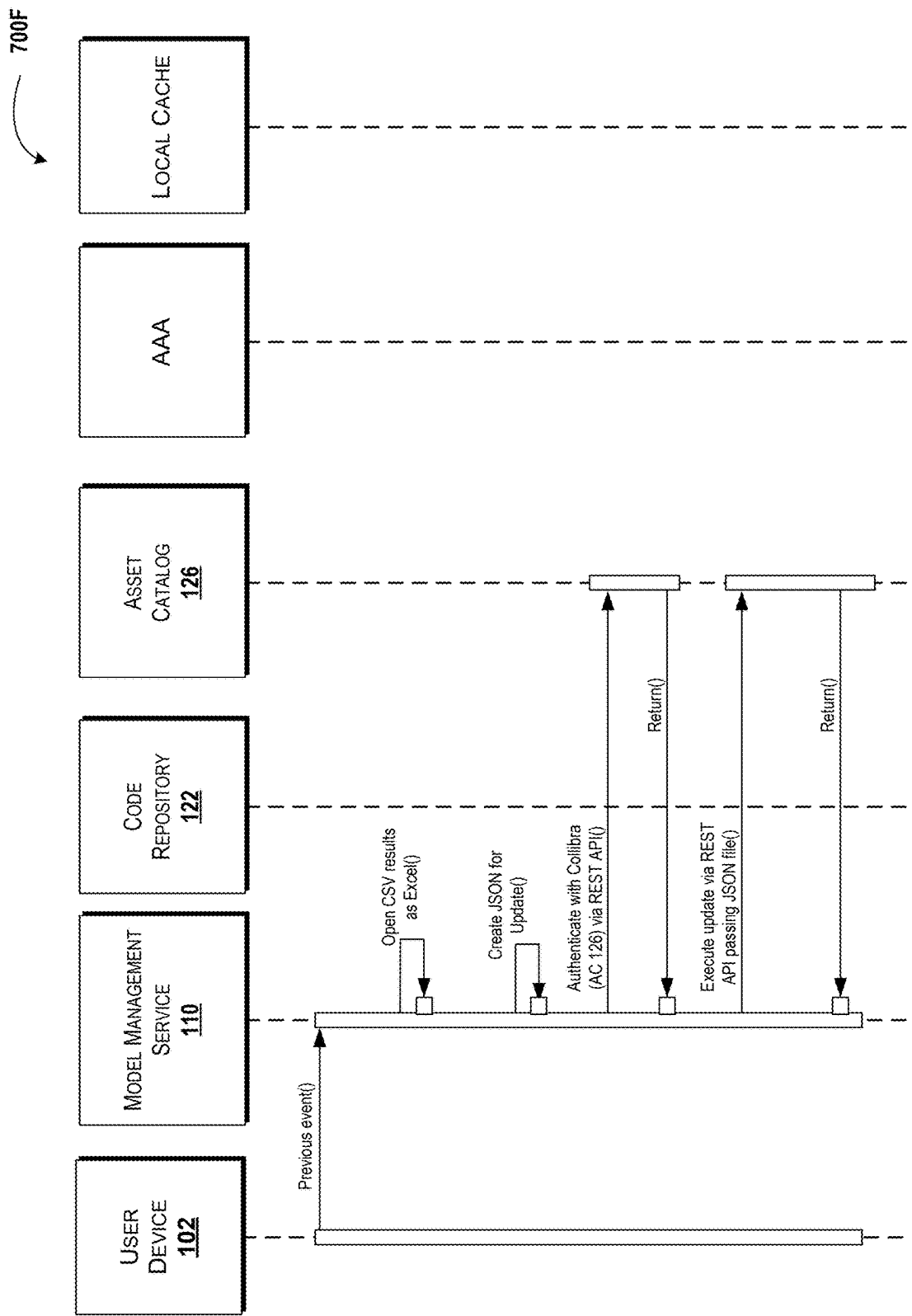


FIG. 7F

700G

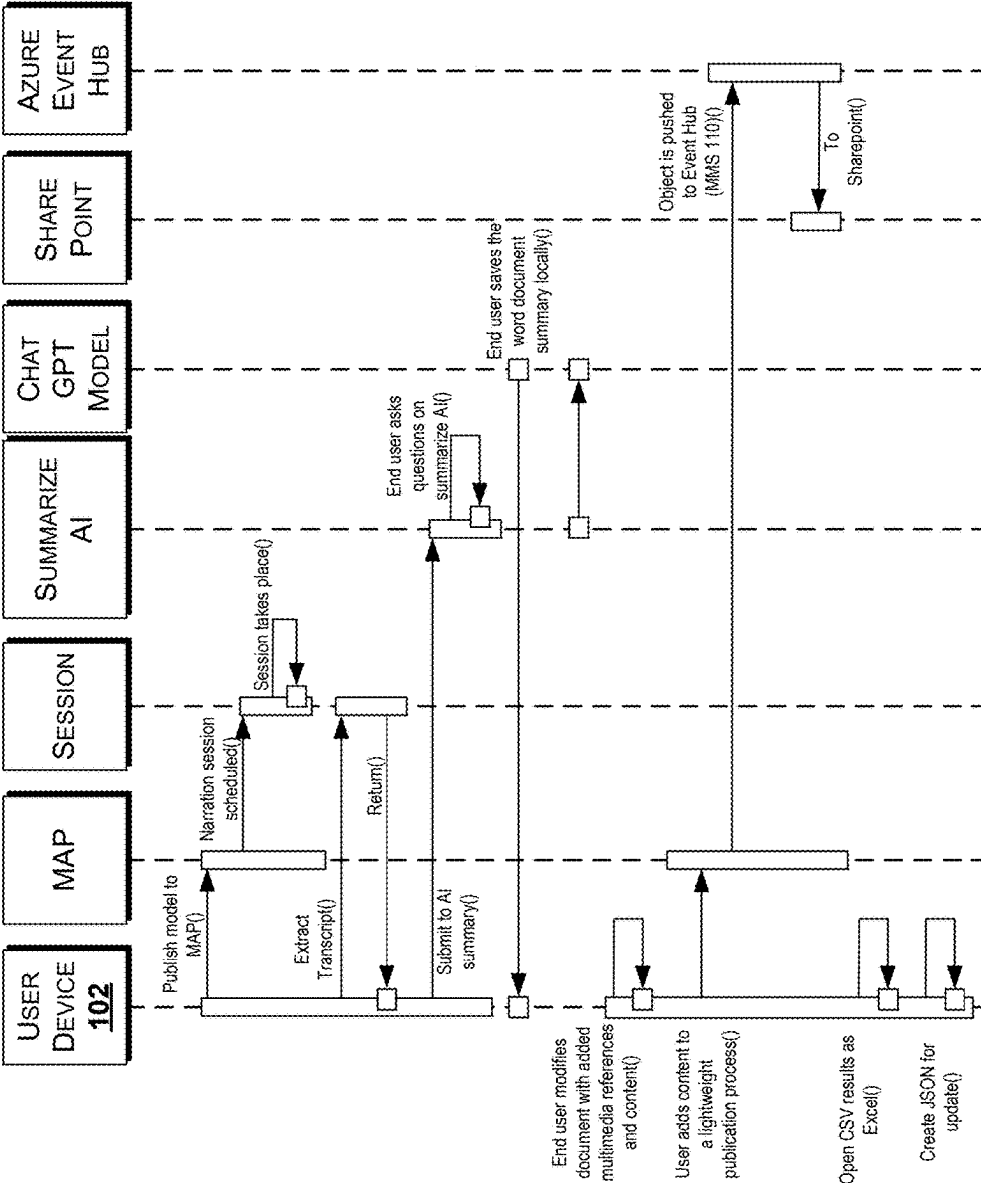
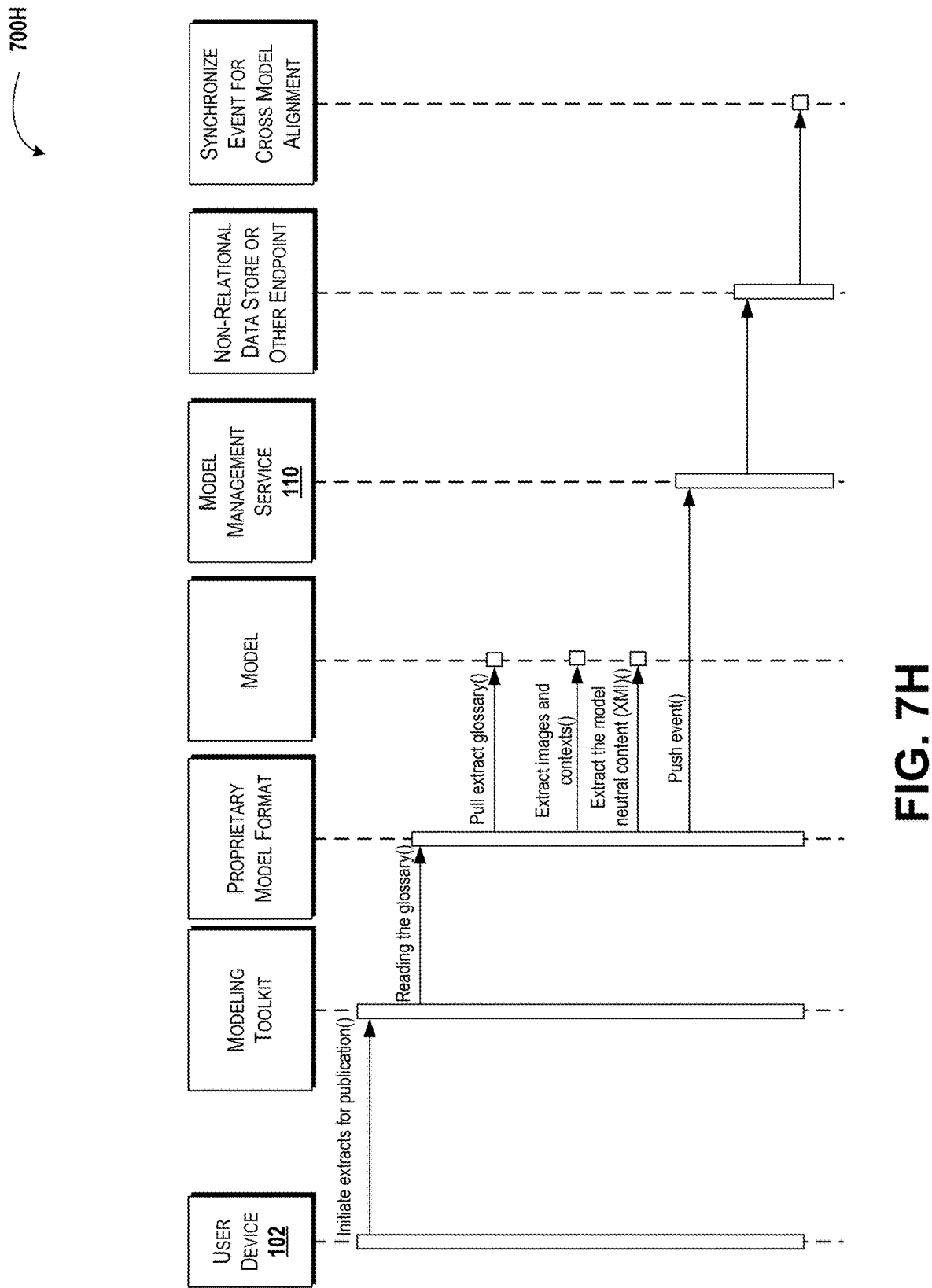


FIG. 7G



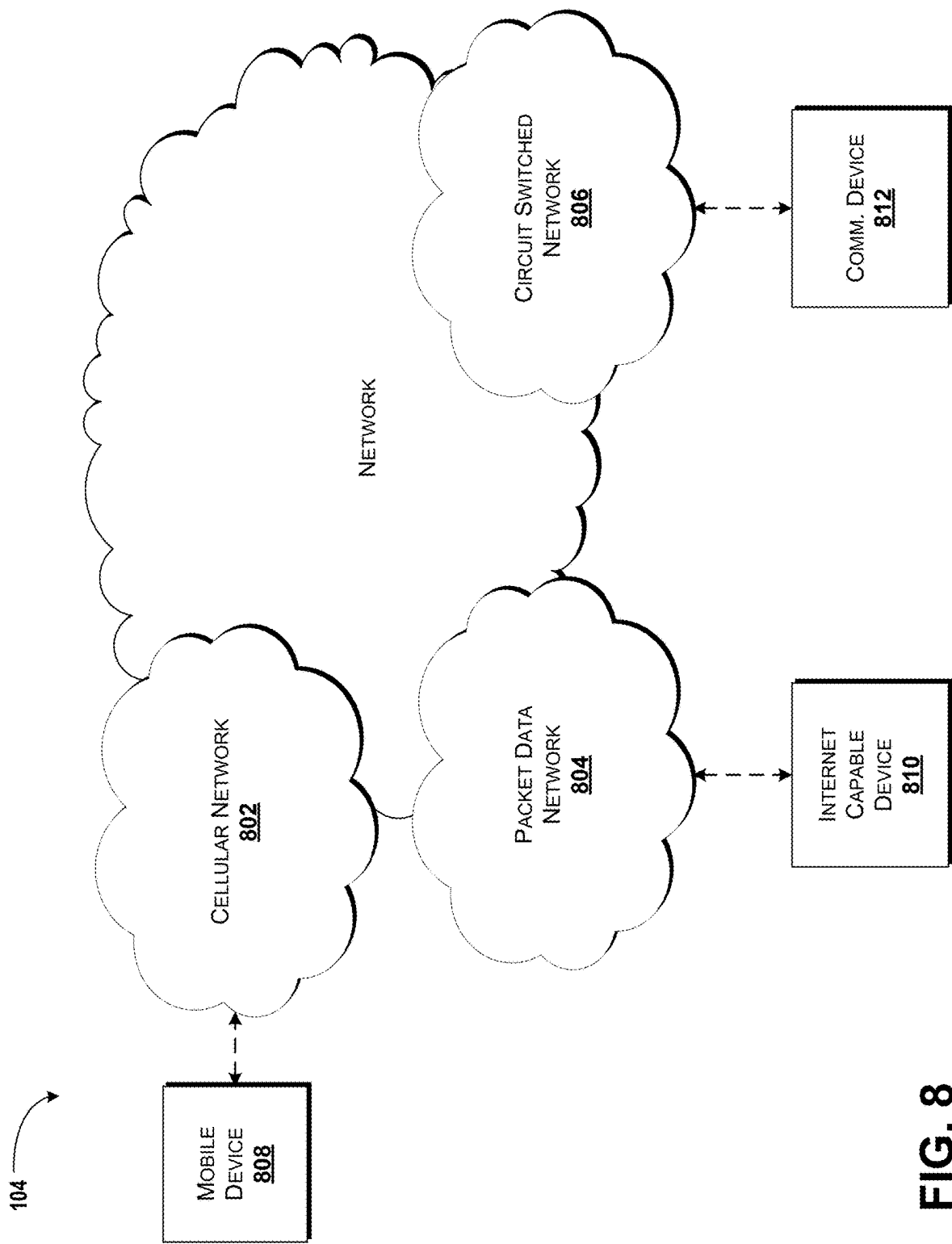


FIG. 8

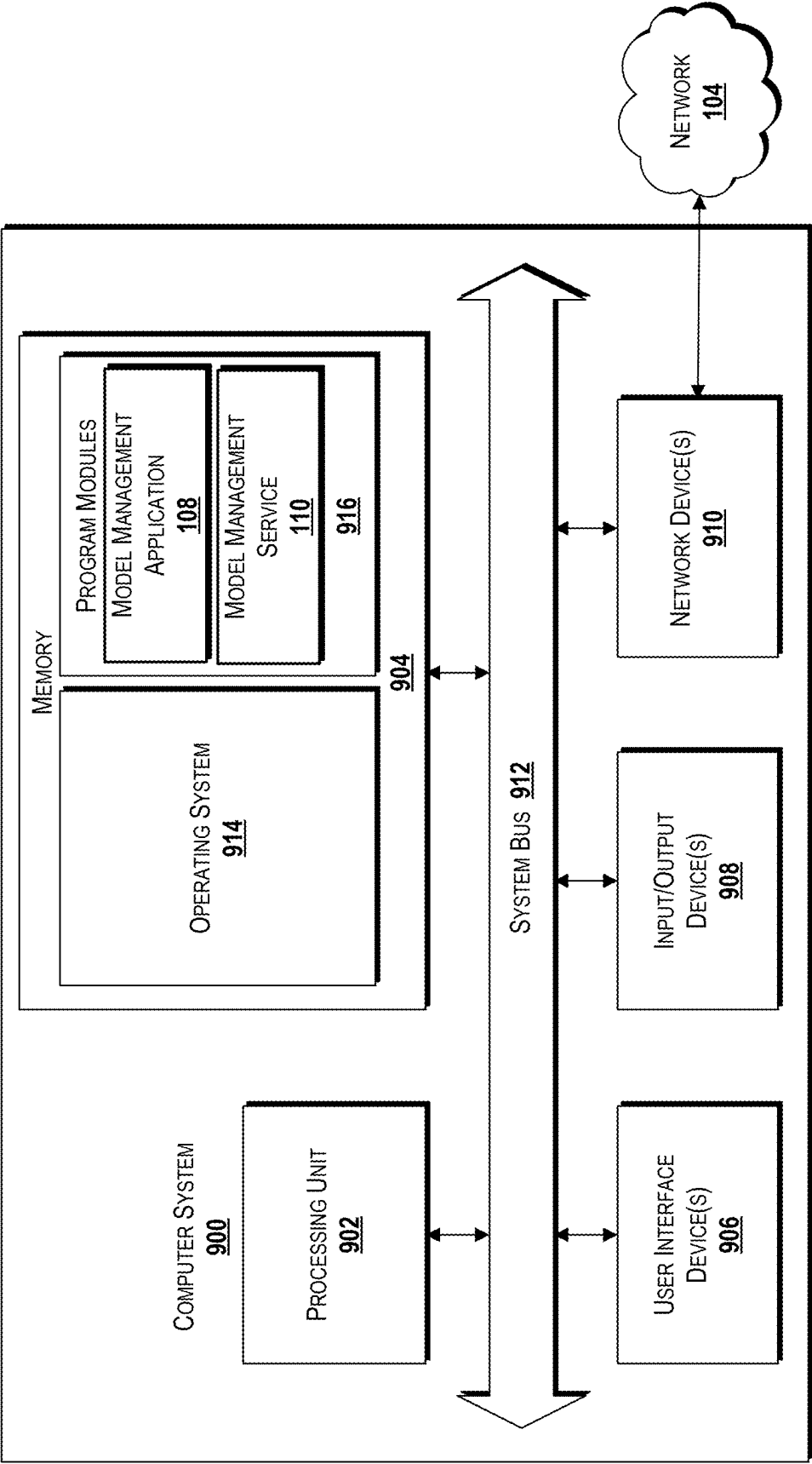


FIG. 9

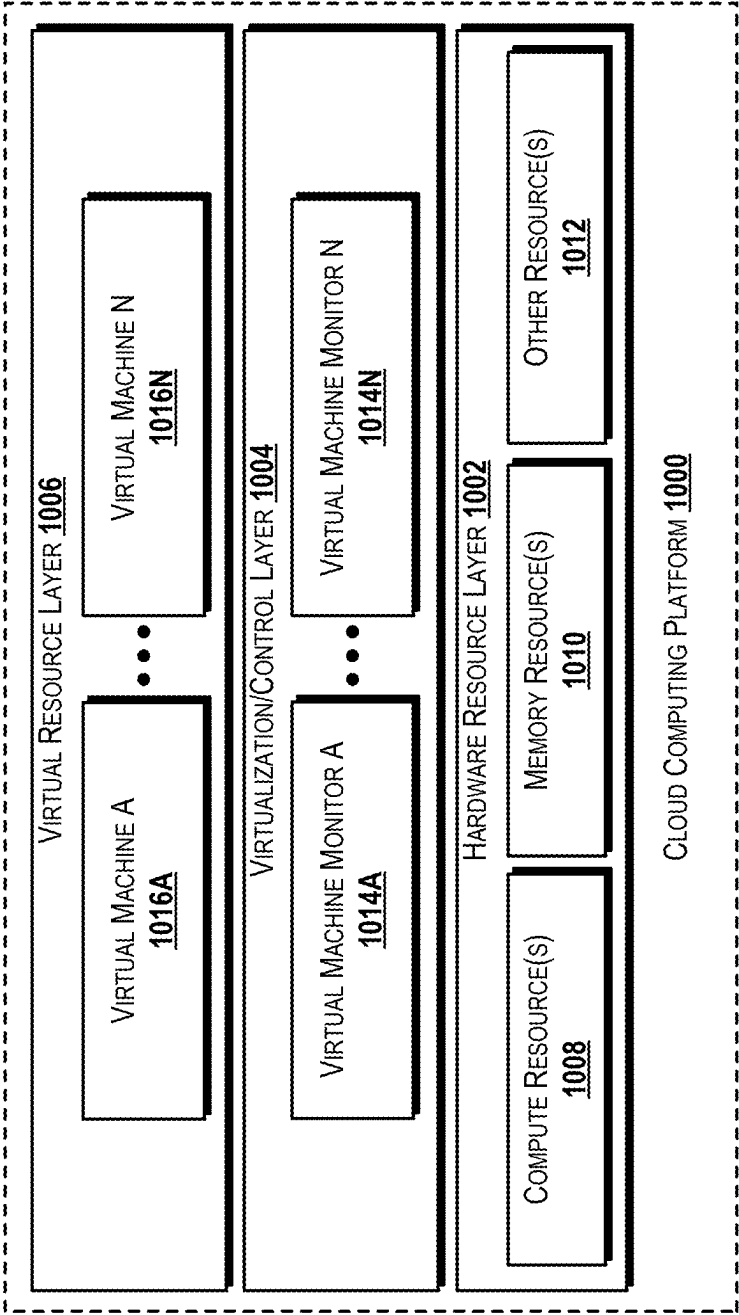


FIG. 10

MODEL CREATION AND MANAGEMENT USING A MODEL MANAGEMENT SERVICE

BACKGROUND

[0001] In the past, various projects (e.g., network evolution of network flows, network devices, network systems, organizations, or the like) would progress via work on the project by a team of individuals who were intimately familiar with the project and the history of the project. With rapid turnover in many organizations, this is not necessarily the case in some circumstances. Thus, valuable information about projects and/or their evolution may be lost and/or may not be accounted for in all circumstances.

SUMMARY

[0002] The present disclosure is directed to model creation and management using a model management service. An integrated information collection mechanism can obtain business metadata from a variety of sources. The integrated information collection mechanism (a model management service) can obtain the metadata and trigger targeted transformation to one or more metadata tracking systems. Modeling in each mapping to common contexts and concepts can enable formerly isolated modeling experts to work with an integrated federated event flow. The model management system also can enable modeling data between different domains and the data between these sources to explore a common view of the modeled information and to allow modeling processes from disparate sources and models to be merged in one area. The independent sources, tools, and domains may all provide input modeling metadata that can be merged in one place, thereby providing a comprehensive and integrated view of the modeled information as well as providing context for the users.

[0003] An event model can synchronize with online storage of metadata from models in catalogs and repositories; distribute information and knowledge by sharing and focusing on understandable linguistics focused on the context of concern; can be configurable and can be used generally for any modeling tool prioritized; and can minimize metadata customization using open standards of metadata and modeling information (including, for example, XMI for XML format, Open Data Standard 1.1 for JSON, or the like) and an optimized network and compute resources using a well-defined event-based approach with rest service standard endpoints.

[0004] A user or other entity (e.g., a user of a user device) can be responsible for modeling projects within an organization and/or entity (e.g., the network, an operator thereof, or the like). A project, as noted above, can include a process, an organization, a network flow, a device, a system, and/or other projects for which modeling (e.g., in system diagrams, UML diagrams, or other forms of models) may be useful. The user device can, for example by execution of the model management application, access a model management service that can be used to create models and/or manage models for the project over time and/or various iterations of the model. In some embodiments, the model management service can access one or more templates (e.g., model templates that can be used to model certain types or categories of projects), a code repository (which can store code for defining models and/or performing operations on other information to generate the models), and an asset catalog

(which can store one or more assets, which can define subprocesses, subsystems, portions of network flows, portions of organizations, and/or other portions of the projects in modeling languages).

[0005] Via accessing the model management service, and/or via the exchange of model management data with the model management service (where the model management data can define the projects being modeled, modeling data, and/or other information), the model management service can generate one or more models. The generated models can be output by the model management service in a proprietary format and/or as neutral format content (e.g., a platform agnostic format such as hypertext markup language (“HTML”), extensible markup language (“XML”), or the like). Thus, the output can include the models and/or model data that can include and/or represent the models, with this output being stored at a cloud-based model repository such as a common repository. It should be understood that this example embodiment is illustrative, and therefore should not be construed as being limiting in any way.

[0006] After creation and storing of the models, the model management service also can support model enrichment, model transformation, model updating, and/or other types of modifications to the model. During these operations (e.g., enrichment, transformation, updating, and/or other modification operations), update events can be tracked by the model management service and used to define model update events and/or other forms of model history events. These update events and iterations of the models can be output to a model history that can be stored at a data storage resource such as the common repository. Thus, a model history can include a history of updates and iterations of the model over the history of updates. It should be understood that this example embodiment is illustrative, and therefore should not be construed as being limiting in any way.

[0007] The model management service also can support providing and supporting interactions between users and/or other entities and a model and model history view. This can enable users or other entities to view models over a lifecycle and/or over lifecycle events. Thus, users can view a history of evolution of a model associated with a project to gain insight into the history of the project and/or to view changes and/or iterations associated with the model. It should be understood that this example embodiment is illustrative, and therefore should not be construed as being limiting in any way.

[0008] According to one aspect of the concepts and technologies disclosed herein, a system is disclosed. The system can include a processor and a memory. The memory can store computer-executable instructions that, when executed by the processor, cause the processor to perform operations. The operations can include obtaining, from an authenticated user device, model management data that can include modeling data that can define a project to model. The project can include a network flow, a network system, a device, or a process. The operations further can include detecting, in the model management data, a selection of a template to apply to the project to generate the model; and creating, based on the model management data and the template, the model. The model can include a directory, a data structure, and a diagram that describes the project. The operations further can include converting the model into a device-agnostic

format version of the model; and loading, to a common repository, model data that can include the device-agnostic format version of the model.

[0009] In some embodiments, the device-agnostic format version of the model can include hypertext markup language or extensible markup language. In some embodiments, the common repository can further store a model history associated with the model, the model history including data that can describe creation of the model. In some embodiments, operations can further include detecting a model update including a request to update the model; retrieving, from the common repository, the model; updating the model based on the model update; generating a model update event that defines an update to the model; adding the model update event to a model history stored at the common repository; and storing an updated model and an updated model history at the common repository.

[0010] In some embodiments, the model update event can include an addition to the model, and updating the model can include obtaining an asset from an asset catalog, the asset relating to the addition; and updating the model based on the asset and the model update event. In some embodiments, operations can further include receiving a request for a model history view relating to the model; obtaining model data and a model history from the common repository, the model data describing the model and the model history describing lifecycle events associated with the model; generating the model history view, the model history view depicting model update events associated with the model and versions of the model; and enabling interactions with the model history view by the authenticated user device via a portal. In some embodiments, the model history view can include depictions of at least two model update events associated with the model, the model update events being obtained from the model history, and the model history view can provide access to at least two versions of the model.

[0011] According to another aspect of the concepts and technologies disclosed herein, a method is disclosed. The method can include obtaining, at a server computer including a processor and from an authenticated user device, model management data that can include modeling data that defines a project to model. The project can include a network flow, a network system, a device, or a process. The method also can include detecting, by the processor and in the model management data, a selection of a template to apply to the project to generate the model; and creating, by the processor and based on the model management data and the template, the model. The model can include a directory, a data structure, and a diagram that describes the project. The method also can include converting, by the processor, the model into a device-agnostic format version of the model; and loading, by the processor and to a common repository, model data that can include the device-agnostic format version of the model.

[0012] In some embodiments, the device-agnostic format version of the model can include hypertext markup language or extensible markup language. In some embodiments, the common repository also stores a model history associated with the model, wherein the model history can include data that describes creation of the model. In some embodiments, the method can further include detecting, by the processor, a model update including a request to update the model; retrieving, by the processor and from the common repository, the model; updating, by the processor, the model based

on the model update; generating, by the processor, a model update event that defines an update to the model; adding, by the processor, the model update event to a model history stored at the common repository; and storing, by the processor, an updated model and an updated model history at the common repository.

[0013] In some embodiments, the model update event can include an addition to the model, and updating the model can include obtaining an asset from an asset catalog, the asset relating to the addition; and updating the model based on the asset and the model update event. In some embodiments, the method can further include receiving a request for a model history view relating to the model; obtaining model data and a model history from the common repository, the model data describing the model and the model history describing lifecycle events associated with the model; and generating the model history view. The model history view can include depictions of at least two model update events associated with the model, the at least two model update events being obtained from the model history. The model history view also can provide access to at least two versions of the model. The method also can include enabling interactions with the model history view by the authenticated user device via portal.

[0014] According to yet another aspect of the concepts and technologies disclosed herein, a computer storage medium is disclosed. The computer storage medium can store computer-executable instructions that, when executed by a processor, cause the processor to perform operations. The operations can include obtaining, from an authenticated user device, model management data that can include modeling data that can define a project to model. The project can include a network flow, a network system, a device, or a process. The operations further can include detecting, in the model management data, a selection of a template to apply to the project to generate the model; and creating, based on the model management data and the template, the model. The model can include a directory, a data structure, and a diagram that describes the project. The operations further can include converting the model into a device-agnostic format version of the model; and loading, to a common repository, model data that can include the device-agnostic format version of the model.

[0015] In some embodiments, the device-agnostic format version of the model can include hypertext markup language or extensible markup language. In some embodiments, the common repository can further store a model history associated with the model, the model history including data that can describe creation of the model. In some embodiments, operations can further include detecting a model update including a request to update the model; retrieving, from the common repository, the model; updating the model based on the model update; generating a model update event that defines an update to the model; adding the model update event to a model history stored at the common repository; and storing an updated model and an updated model history at the common repository.

[0016] In some embodiments, the model update event can include an addition to the model, and updating the model can include obtaining an asset from an asset catalog, the asset relating to the addition; and updating the model based on the asset and the model update event. In some embodiments, operations can further include receiving a request for a model history view relating to the model; obtaining model

data and a model history from the common repository, the model data describing the model and the model history describing lifecycle events associated with the model; generating the model history view, the model history view depicting model update events associated with the model and versions of the model; and enabling interactions with the model history view by the authenticated user device via a portal. In some embodiments, the model history view can include depictions of at least two model update events associated with the model, the model update events being obtained from the model history, and the model history view can provide access to at least two versions of the model.

[0017] Other systems, methods, and/or computer program products according to embodiments will be or become apparent to one with skill in the art upon review of the following drawings and detailed description. It is intended that all such additional systems, methods, and/or computer program products be included within this description and be within the scope of this disclosure.

BRIEF DESCRIPTION OF THE DRAWINGS

[0018] FIG. 1 is a system diagram illustrating an illustrative operating environment for various embodiments of the concepts and technologies described herein.

[0019] FIG. 2 is a flow diagram showing aspects of a method for creating, storing, modifying, and interacting with a model using a model management service, according to an illustrative embodiment of the concepts and technologies described herein.

[0020] FIG. 3A is a flow diagram showing aspects of a method for creating and storing a model using a model management service, according to an illustrative embodiment of the concepts and technologies described herein.

[0021] FIG. 3B is a flow diagram showing aspects of a method for creating and storing a model using a model management service, according to another illustrative embodiment of the concepts and technologies described herein.

[0022] FIG. 4 is a flow diagram showing aspects of a method for enriching, transforming, and updating a model using a model management service, according to an illustrative embodiment of the concepts and technologies described herein.

[0023] FIG. 5 is a flow diagram showing aspects of a method for viewing and interacting with a model history using a model management service, according to an illustrative embodiment of the concepts and technologies described herein.

[0024] FIGS. 6A-6C are user interface diagrams showing various screen displays showing interacting with a model history view, according to some illustrative embodiments of the concepts and technologies described herein.

[0025] FIGS. 7A-7H are example call flow diagrams illustrating additional aspects of the concepts and technologies disclosed herein for creating, storing modifying, and interacting with a model using a model management service, according to some illustrative embodiments of the concepts and technologies described herein.

[0026] FIG. 8 schematically illustrates a network, according to an illustrative embodiment of the concepts and technologies described herein.

[0027] FIG. 9 is a block diagram illustrating an example computer system configured to provide model creation and

management using a model management service, according to some illustrative embodiments of the concepts and technologies described herein.

[0028] FIG. 10 is a diagram illustrating a computing environment capable of implementing aspects of the concepts and technologies disclosed herein, according to some illustrative embodiments of the concepts and technologies described herein.

DETAILED DESCRIPTION

[0029] The following detailed description is directed to providing model creation and management using a model management service. A user or other entity (e.g., a user of a user device) can be responsible for modeling projects within an organization and/or entity (e.g., the network, an operator thereof, or the like). A project, as noted above, can include a process, an organization, a network flow, a device, a system, and/or other projects for which modeling (e.g., in system diagrams, UML diagrams, or other forms of models) may be useful. The user device can, for example by execution of the model management application, access a model management service that can be used to create models and/or manage models for the project over time and/or various iterations of the model. In some embodiments, the model management service can access one or more templates (e.g., model templates that can be used to model certain types or categories of projects), a code repository (which can store code for defining models and/or performing operations on other information to generate the models), and an asset catalog (which can store one or more assets, which can define subprocesses, subsystems, portions of network flows, portions of organizations, and/or other portions of the projects in modeling languages).

[0030] Via accessing the model management service, and/or via the exchange of model management data with the model management service (where the model management data can define the projects being modeled, modeling data, and/or other information), the model management service can generate one or more models. The generated models can be output by the model management service in a proprietary format and/or as neutral format content (e.g., a platform agnostic format such as HTML, XML, or the like). Thus, the output can include the models and/or model data that can include and/or represent the models, with this output being stored at a cloud-based model repository such as a common repository. It should be understood that this example embodiment is illustrative, and therefore should not be construed as being limiting in any way.

[0031] After creation and storing of the models, the model management service also can support model enrichment, model transformation, model updating, and/or other types of modifications to the model. During these operations (e.g., enrichment, transformation, updating, and/or other modification operations), update events can be tracked by the model management service and used to define model update events and/or other forms of model history events. These update events and iterations of the models can be output to a model history that can be stored at a data storage resource such as the common repository. Thus, a model history can include a history of updates and iterations of the model over the history of updates. It should be understood that this example embodiment is illustrative, and therefore should not be construed as being limiting in any way.

[0032] The model management service also can support providing and supporting interactions between users and/or other entities and a model and model history view. This can enable users or other entities to view models over a lifecycle and/or over lifecycle events. Thus, users can view a history of evolution of a model associated with a project to gain insight into the history of the project and/or to view changes and/or iterations associated with the model. It should be understood that this example embodiment is illustrative, and therefore should not be construed as being limiting in any way.

[0033] While the subject matter described herein is presented in the general context of program modules that execute in conjunction with the execution of an operating system and application programs on a computer system, those skilled in the art will recognize that other implementations may be performed in combination with other types of program modules. Generally, program modules include routines, programs, components, data structures, and other types of structures that perform particular tasks or implement particular abstract data types. Moreover, those skilled in the art will appreciate that the subject matter described herein may be practiced with other computer system configurations, including hand-held devices, multiprocessor systems, microprocessor-based or programmable consumer electronics, minicomputers, mainframe computers, and the like.

[0034] Referring now to FIG. 1, aspects of an operating environment 100 for various embodiments of the concepts and technologies disclosed herein for providing model creation and management using a model management service will be described, according to an illustrative embodiment. The operating environment 100 shown in FIG. 1 includes one or more user device 102A-N (hereinafter collectively and/or generically referred to as the “user devices 102”). The user devices 102 can operate in communication with and/or as part of a communications network (“network”) 104, though this is not necessarily the case in all embodiments. The user device 102A will be described in more detail for purposes of illustrating the concepts and technologies disclosed herein, but it should be understood that one or more (or even all) of the user devices 102 may have an architecture similar or even identical to the user device 102A and/or that one or more (or even all) of the user devices 102 may perform the functionality illustrated and described herein with respect to the user device 102A. As such, it should be understood that the illustrated and described example embodiment is illustrative, and therefore should not be construed as being limiting in any way.

[0035] According to various embodiments, the functionality of user device 102A may be provided by one or more server computers, one or more desktop computers, one or more mobile telephones, one or more smartphones, one or more tablet computers, one or more laptop computers, other computing systems, and the like. It should be understood that the functionality of user device 102A may be provided by a single device, by two or more similar devices, and/or by two or more dissimilar devices. For purposes of describing the concepts and technologies disclosed herein, the user device 102A is described herein as a personal computer such as a laptop computer, a desktop computer, a tablet computer, or the like. It should be understood that this embodiment is illustrative, and should not be construed as being limiting in any way.

[0036] The user device 102A can execute an operating system 106 and one or more application programs such as, for example, a model management application 108. The operating system 106 can include a computer program that can control the operation of the user device 102. The model management application 108 can include an executable program that can be configured to execute on top of the operating system 106 to provide various functions as illustrated and described herein. In some embodiments of the concepts and technologies disclosed herein, as will be appreciated with reference to the description herein, the functionality of the model management application 108 may be provided in some embodiments by a web browser or the like, which may provide various functions as illustrated and described by way of interacting with a model management service 110 executed and/or hosted on a server computer 112. The model management application 108 may provide the functionality in some embodiments, for example, by rendering and/or exchanging model management data 114, which may be exchanged by the user device 102A and the server computer 112 via communications between the user device 102A and the server computer 112 (e.g., in some embodiments via data exchanges via a portal or API exposed by the server computer 112). It should be understood that this example embodiment is illustrative, and therefore should not be construed as being limiting in any way.

[0037] According to various embodiments of the concepts and technologies disclosed herein, the model management data 114 can include, for example, authentication data, commands and/or requests, modeling data, model updates, scripts, and/or other data. The authentication data can include data and/or data communications that may be exchanged by the user device 102A and the server computer 112 (or other device such as an authentication server or the like) to authenticate the user device 102A as a user of the model management service 110. In various embodiments of the concepts and technologies disclosed herein, the authentication data can include logins, passwords, network identifiers, and/or other information that can be used to determine if the user device 102A and/or a user thereof is permitted to access and/or use the functionality of the model management service 110. Thus, it can be appreciated that the model management service 110 can be a subscription-based service and/or a service that is limited to certain users, devices, and/or organizations in various embodiments. It should be understood that this example embodiment is illustrative, and therefore should not be construed as being limiting in any way.

[0038] The commands and/or requests can include, for example, requests to authenticate (e.g., from the user device 102A and/or other user devices 102), requests for providing challenges (e.g., logins, passwords, or the like), requests for types of data and/or information associated with the modeling functionality of the model management service 110 illustrated and described herein, and/or other requests and/or responses to the requests as illustrated and described herein. The commands and requests also can include commands to trigger various actions as illustrated and described herein or the like. It should be understood that these example embodiments are illustrative, and therefore should not be construed as being limiting in any way.

[0039] The modeling data can include various types of information that can define representations of processes and/or devices to the model management service 110. Thus,

the modeling data can include data that describes processes, devices, or the like. The modeling data can be used by the model management service 110 to generate one or more models 116 as will be explained in more detail herein. The model updates can include changes to the models 116 created by the model management service 110. For example, the model updates can be used to transform the models 116; to introduce functionality to the models 116; to iterate versions of the models 116; to save histories relating to models 116, their evolution, and/or changes thereto; combinations thereof; or the like. Because other model updates are possible and are contemplated as illustrated and described herein, it should be understood that these example embodiments are illustrative, and therefore should not be construed as being limiting in any way.

[0040] The scripts can include executable code (e.g., JSON files, or the like), which can be loaded to the user devices 102 and/or executed thereby (e.g., a JSON file that can be provided to the user device 102A and executed within a browser environment as is generally understood). Thus, the scripts can include applications, network scripts, and/or other executable files that can be loaded to the user device 102A and executed thereby. It should be understood that this example embodiment is illustrative, and therefore should not be construed as being limiting in any way.

[0041] The other data can include other types of information illustrated and described herein such as, for example, location information, permission lists, graphics data (e.g., renderable displays), other types of information, combinations thereof, or the like. The model management data 114 can be created by the user devices 102 and/or the server computer 112, as can be appreciated by referring to the description of the model management service 110 hereinbelow.

[0042] According to various embodiments of the concepts and technologies disclosed herein, the model management service 110 can be configured to communicate with one or more of the user devices 102 (e.g., via exchange of the model management data 114 with the model management application 108) to provide the various model lifecycle management illustrated and described herein. According to some embodiments, the model management service 110 can communicate with the user devices 102 to enable users and/or other authorized users of the user devices 102 to create one or more model(s) 116 (e.g., via one or more model templates (hereinafter referred to and illustrated in FIG. 1 as “templates 118”) in some embodiments and/or via other functionality); to store the models 116 in one or more data storage devices and/or entities in communication with the server computer 112; to enable one or more user devices 102 to enrich, update, and/or transform the models 116 (e.g., via obtaining and/or saving code 120 from and/or to one or more code repository 122; via obtaining and/or using one or more assets 124 from one or more asset catalog 126); and/or via other operations as will be illustrated and described herein in detail below. The model management service 110 also can be configured to create and/or generate model data 128, which can include model updates and/or enrichments, models 116 in a proprietary format or represented in neutral format content (e.g., HTML XML, or other forms of neutral format content); one or more model history events (e.g., events that mark and/or define creation, evolution, and/or enrichment events relating to one or more of the models

116); and/or other data such as e.g., transcripts, summaries, tables, scripts, CSV files, and the like.

[0043] In some embodiments, the model data 128 can be stored by the server computer 112 in one or more data repositories such as the common repository 130 illustrated in FIG. 1. The functionality of the common repository 130 can be provided by one or more databases, server computers, desktop computers, laptop computers, other computing systems, and the like. In the illustrated embodiments, the functionality of the common repository can be provided by a cloud-based data storage platform. It should be understood that this example embodiment is illustrative, and therefore should not be construed as being limiting in any way.

[0044] It can be appreciated from the above that the model management service 110 can be configured to enable viewing of models 116 in one or more renderable displays; to enable viewing of one or more history of the models 116 representing creation, evolution, and/or enrichment of the models 116 (hereinafter collectively and/or generically referred to as “model histories 132”); and/or to provide other functionality as illustrated and described hereinbelow. It should be understood that this example embodiment is illustrative, and therefore should not be construed as being limiting in any way.

[0045] According to some embodiments of the concepts and technologies disclosed herein, the model management service 110 can be configured to support a model creation and management lifecycle for various users and/or entities. At a high level, the lifecycle of a model 116 can include a creation phase, wherein a user or other entity can create one or more models 116 (optionally using one or more templates 118 in some embodiments) and save the models 116 to a data storage location (e.g., a local cache or data storage device, a common repository in communication with the user devices 102 and/or the server computer 112, and/or other repositories or data storage resources); a model enrichment, updating, and/or transformation phase wherein one or more entities (e.g., one or more user device 102) can access models 116 and edit, update, and/or otherwise modify the models 116; and a model 116 and/or model history 132 viewing phase wherein one or more entities (e.g., one or more user devices 102) can access one or more iterations of the models 116 and/or model histories 132. Of course, other operations as illustrated and described herein can be performed by the model management service 110 during the lifecycle of the models 116. As such, it should be understood that this example embodiment is illustrative, and therefore should not be construed as being limiting in any way.

[0046] In some embodiments, a model creation phase as mentioned above can include a number of operations that can be provided by the model management service 110 and/or the model management application 108. As noted above, the model management application 108 can be configured to interact with the model management service 110 to perform the various operations illustrated and described herein with reference to the user devices 102 and in some instances can include a natively-executed stand-alone application and/or other types of applications such as a web browser (which may be configured to obtain model management data 114 from the model management service 110, e.g., via portal or API, and to render the information, e.g., in one or more webpages to enable to functionality illustrated and described herein). Because other methods of interacting with the model management service 110 are possible and are

contemplated, it should be understood that this example embodiment is illustrative, and therefore should not be construed as being limiting in any way.

[0047] Before initiating the creation phase of the model management lifecycle, some embodiments of the model management service **110** can enforce an authentication scheme against users and/or devices (e.g., the user device **102A**) attempting to access the model management service **110**. Thus, although not shown in FIG. 1, it can be appreciated that a “triple-A” server (i.e., an AAA server, which generally refers to a server that provides authentication, authorization and accounting services for various entities in a system such as the network **104**) can be provided in some embodiments to manage this authentication process. In some other embodiments, the server computer **112** can include an authentication module or other functionality for providing these and/or other authentication services for the model management service **110**.

[0048] An authenticated user or device (e.g., the user device **102A**) can request, from the model management service **110**, code **120** associated with a project that is to be modeled. The model management service **110** can be configured to obtain code **120** associated with the project from one or more code repositories **122** or other data storage resources. The model management service **110** can provide the code **120** (e.g., as part of the model management data **114** and via a portal, API, webpage, or the like) to the user device **102A** for analysis and/or viewing at the user device **102A**. The user device **102A** also can obtain (e.g., as part of the model management data **114** from the server computer **112** or other entities) one or more scripts (e.g., network scripts) such as JSON files, or the like, for execution at the user device **102A**.

[0049] The scripts can be executed by the user device **102A** (e.g., via execution of the model management application **108**) to create a model **116** based on the project and/or other data such as the code **120** and/or scripts. During creation of the model **116**, the user device **102A** or other entity (e.g., the server computer **112**) can generate one or more directories, one or more data structures, and/or one or more diagrams that collectively and/or individually can be considered the models **116** illustrated and described herein. The user device **102A** can be configured to load the models **116** to the common repository **130** in some embodiments, or to provide, to the server computer **112**, modeling data to the server computer **112** and the server computer **112** can store the models **116** at the common repository **130** or elsewhere. It should be understood that this example embodiment is illustrative, and therefore should not be construed as being limiting in any way.

[0050] After creation of the models **116**, one or more users, devices, or other entities can be configured to update the models **116**. In particular, various operations can be performed on the models **116** to transform the models **116**, to update the models **116**, to record history events associated with the models **116**, and/or otherwise to modify the models **116** as illustrated and described herein. According to various embodiments of the concepts and technologies disclosed herein, the models **116** can be updated for any of a variety of reasons, some of which are described herein as example embodiments.

[0051] For example, in one contemplated embodiment, a user or other entity (e.g., a user associated with the user device **102A**) may update a job aid. This change may result

in a change to one or more processes associated with a particular model **116**. For example, if a model **116** represents a network flow, network architecture, network flow diagram, or the like, a change to a job aid may result in a change to the network flow, the network architecture, the network flow diagram, or the like. It should be understood that this example embodiment is illustrative, and therefore should not be construed as being limiting in any way. Other contemplated embodiments illustrated and described herein include a transformation request for the model **116** (e.g., adding or removing assets, changing the model **116**, or the like) and recording events associated with the models **116**. It should be understood that these example embodiments are illustrative, and therefore should not be construed as being limiting in any way.

[0052] In response to the model management service **110** detecting a model update event such as those mentioned above (e.g., a transformation request, a job aid update, an event, or the like), the model management service **110** can be configured to update the model **116** based on the model update event. The model management service **110** also can be configured to propagate the changes to other entities and/or to trigger updates to other entities (e.g., to update across platforms, to update assets **124** in the asset catalog **126**, the update code **120** in the code repository **122**, or the like). The model management service **110** can store the updated model **116** and/or a model history **132** that can track the (now at least two) versions of the models **116** including the original model **116** and the updated version of the model **116**. The model management service **110** can generate model data **128** (which can represent the model updates and model history events) and provide the updated model and model history **132** (or updated model history **132**) to the common repository **130** or other data storage resource. In various embodiments of the concepts and technologies disclosed herein, the model data **128** also can represent the model **116** as neutral format content (e.g., HTML content, XML, or the like) so that the model **116** can be viewed across platforms and/or by almost any user or device. It should be understood that this example embodiment is illustrative, and therefore should not be construed as being limiting in any way.

[0053] The model management service **110** also can be configured to enable devices and/or other entities to view models **116** and model histories **132**, as well as interacting with the models **116** and/or model histories **132**. In some example embodiments, the model management service **110** can receive a request to view a model **116** and/or a model history view (e.g., a view of a model history **132** with links to iterations of models **116** and/or related files such as diagrams, structures, code **120**, and the like). In response to the request, the model management service **110** can be configured to identify the model **116** (associated with the requested model **116** and/or requested model history **132**) and obtain model data **128** associated with the identified model **116** and/or model history **132**. In various embodiments of the concepts and technologies disclosed herein, the model management service **110** can obtain the model data **128** from the common repository **130** and/or other data storage resources or structures.

[0054] The model management service **110** can be configured to generate, based on the model data **128**, a model history view, which can include renderable data that can be provided to one or more users, devices, or entities (e.g., the user devices **102**) for viewing. As noted above, the model

history views (some examples of which are illustrated and described below with reference to FIGS. 6A-6C) can provide a graphical representation of the model history 132 and one or more models 116 represented in the model history 132. Thus, users can see an evolution of a model 116 during its life cycle from creation to a most recent iteration, and can be enabled to access associated models 116 and/or related data such as diagrams, structures, or the like across the life cycle events. The model management service 110 can enable interactions with the model history view and provide updated iterations of the model history view at appropriate times in various embodiments. It should be understood that this example embodiment is illustrative, and therefore should not be construed as being limiting in any way.

[0055] According to various embodiments of the concepts and technologies disclosed herein, the model management service 110 can also be configured to collect one or more streams of metadata 134 relating to one or more projects and/or models 116 of the projects. The metadata 134 can be collected and/or otherwise obtained from any number of network systems, devices, software, or the like (hereinafter referred to as “metadata sources 136”). In various embodiments of the concepts and technologies disclosed herein, the model management service 110 can detect model update events based on the metadata 134. Thus, the models 116 can be based on any metadata 134 relating to a project and/or model 116, and the model data 128 can represent the metadata 134 in various embodiments. It should be understood that this example embodiment is illustrative, and therefore should not be construed as being limiting in any way.

[0056] In general, a user or other entity (e.g., a user of a user device 102A) can be responsible for modeling projects within an organization and/or entity (e.g., the network 104, an operator thereof, or the like). A project, as noted above, can include a process, an organization, a network flow, a device, a system, and/or other projects for which modeling (e.g., in system diagrams, UML diagrams, or other forms of models) may be useful. The user device 102A can, for example by execution of the model management application 108, access a model management service 110 that can be used to create models and/or manage models 116 for the project over time and/or various iterations of the model 116. In some embodiments, the model management service 110 can access one or more templates 118 (e.g., model templates that can be used to model certain types or categories of projects), a code repository 122 (which can store code 120 for defining models 116 and/or performing operations on other information to generate the models 116), and an asset catalog 126 (which can store one or more assets 124, which can define subprocesses, subsystems, portions of network flows, portions of organizations, and/or other portions of the projects in modeling languages).

[0057] Via accessing the model management service 110, and/or via the exchange of model management data 114 with the model management service 110 (where the model management data 114 can define the projects being modeled, modeling data, and/or other information), the model management service 110 can generate one or more models 116. The generated models 116 can be output by the model management service 110 in a proprietary format and/or as neutral format content (e.g., a platform agnostic format such as HTML, XML, or the like). Thus, the output can include the models 116 and/or model data 128 that can include

and/or represent the models 116, with this output being stored at a cloud-based model repository such as the common repository 130 illustrated in FIG. 1. It should be understood that this example embodiment is illustrative, and therefore should not be construed as being limiting in any way.

[0058] After creation and storing of the models 116, the model management service 110 also can support model enrichment, model transformation, model updating, and/or other types of modifications to the model 116. During these operations (e.g., transformation, updating, and/or other modification operations), update events can be tracked by the model management service 110 and used to define model update events and/or other forms of model history events. These update events and iterations of the models 116 can be output to a model history 132 that can be stored at a data storage resource such as the common repository 130. Thus, a model history 132 can include a history of updates and iterations of the model 116 over the history of updates. It should be understood that this example embodiment is illustrative, and therefore should not be construed as being limiting in any way.

[0059] The model management service 110 also can support providing and supporting interactions between users and/or other entities and a model and model history view. This can enable users or other entities to view models 116 over a lifecycle and/or over lifecycle events. Thus, users can view a history of evolution of a model 116 associated with a project to gain insight into the history of the project and/or to view changes and/or iterations associated with the model 116. It should be understood that this example embodiment is illustrative, and therefore should not be construed as being limiting in any way.

[0060] FIG. 1 illustrates four user devices 102, one network 104, one server computer 112, one code repository 122, one asset catalog 126, and one common repository 130. It should be understood, however, that various implementations of the operating environment 100 can include zero, one, two, three, four, or more than four user devices 102; one or more than one network 104; zero, one, or more than one server computer 112; zero, one, or more than one code repository 122; zero, one, or more than one asset catalog 126; and zero, one, or more than one common repository 130. As such, the illustrated embodiment should be understood as being illustrative, and should not be construed as being limiting in any way.

[0061] Turning now to FIG. 2, aspects of a method 200 for creating, storing, modifying, and interacting with a model using a model management service will be described in detail, according to an illustrative embodiment. It should be understood that the operations of the methods disclosed herein are not necessarily presented in any particular order and that performance of some or all of the operations in an alternative order(s) is possible and is contemplated. The operations have been presented in the demonstrated order for ease of description and illustration. Operations may be added, omitted, and/or performed simultaneously, without departing from the scope of the concepts and technologies disclosed herein.

[0062] It also should be understood that the methods disclosed herein can be ended at any time and need not be performed in its entirety. Some or all operations of the methods, and/or substantially equivalent operations, can be performed by execution of computer-readable instructions

included on a computer storage media, as defined herein. The term “computer-readable instructions,” and variants thereof, as used herein, is used expansively to include routines, applications, application modules, program modules, programs, components, data structures, algorithms, and the like. Computer-readable instructions can be implemented on various system configurations including single-processor or multiprocessor systems, minicomputers, main-frame computers, personal computers, hand-held computing devices, microprocessor-based, programmable consumer electronics, combinations thereof, and the like.

[0063] Thus, it should be appreciated that the logical operations described herein are implemented (1) as a sequence of computer implemented acts or program modules running on a computing system and/or (2) as interconnected machine logic circuits or circuit modules within the computing system. The implementation is a matter of choice dependent on the performance and other requirements of the computing system. Accordingly, the logical operations described herein are referred to variously as states, operations, structural devices, acts, or modules. These states, operations, structural devices, acts, and modules may be implemented in software, in firmware, in special purpose digital logic, and any combination thereof. As used herein, the phrase “cause a processor to perform operations” and variants thereof is used to refer to causing a processor of a computing system or device, such as the user device 102 and/or the server computer 112, to perform one or more operations and/or causing the processor to direct other components of the computing system or device to perform one or more of the operations.

[0064] For purposes of illustrating and describing the concepts of the present disclosure, the method 200 is described herein as being performed by the server computer 112 via execution of one or more software modules such as, for example, the model management service 110. It should be understood that additional and/or alternative devices and/or network nodes can provide the functionality described herein via execution of one or more modules, applications, and/or other software including, but not limited to, the model management service 110. Thus, the illustrated embodiments are illustrative, and should not be viewed as being limiting in any way.

[0065] The method 200 begins at operation 202. At operation 202, the server computer 112 can be used to create and store one or more models 116. For example, operation 202 can correspond to the server computer 112 enabling the creation of one or more models 116 and the loading of the one or more models 116 to a common repository 130 or other data storage resource. It should be understood that this example embodiment is illustrative, and therefore should not be construed as being limiting in any way.

[0066] In some embodiments of the concepts and technologies disclosed herein, creation of one or more models 116 can include a number of operations that can be provided by the model management service 110 and/or the model management application 108. As mentioned above, some embodiments of the model management service 110 can enforce an authentication requirement for users and/or devices attempting to create the models 116. The model management service 110 can be configured to trigger authentication by other entities (e.g., an authentication server or module), or to include an authentication module or other functionality for providing these and/or other authentication services for the model management service 110.

An authenticated user or device can request, from the model management service 110, code 120 associated with a project that is to be modeled. The model management service 110 can provide the code 120 (e.g., as part of the model management data 114 and via a portal, API, webpage, or the like) to the user device 102A for analysis and/or viewing at the user device 102A. The user device 102A also can obtain (e.g., as part of the model management data 114 from the server computer 112 or other entities) one or more scripts for execution at the user device 102A.

[0067] The scripts can be executed by the user device 102A (e.g., via execution of the model management application 108) to create a model 116 based on the project, one or more templates 118, and/or other data such as the code 120 and/or scripts. During creation of the model 116, the user device 102A or other entity (e.g., the server computer 112) can generate one or more directories, one or more data structures, and/or one or more diagrams that collectively and/or individually can be considered the models 116 illustrated and described herein. The user device 102A can be configured to load the models 116 to a data storage resource (e.g., the common repository 130 in some embodiments), or to provide, to the server computer 112, modeling data to the server computer 112 and the server computer 112 can store the models 116 at the common repository 130 or elsewhere. It should be understood that this example embodiment is illustrative, and therefore should not be construed as being limiting in any way. Additional details of the model creation will be illustrated and described in more detail below, particularly with reference to FIG. 3.

[0068] From operation 202, the method 200 can proceed to operation 204. At operation 204, the server computer 112 can enable model enrichment, transformation, and/or updating. In various embodiments, one or more users, devices, or other entities can be configured to update the models 116. In particular, various operations can be performed on the models 116 to transform the models 116, to update the models 116, to record history events associated with the models 116, and/or otherwise to modify the models 116 as illustrated and described herein. According to various embodiments of the concepts and technologies disclosed herein, the models 116 can be updated for any of a variety of reasons, some of which are described herein as example embodiments.

[0069] In response to the model management service 110 detecting a model update event (e.g., a transformation request, a job aid update, an event, or the like), the model management service 110 can be configured to update the model 116 based on the model update event. The model management service 110 also can be configured to propagate the changes to other entities and/or to trigger updates to other entities (e.g., to update across platforms, to update assets 124 in the asset catalog 126, the update code 120 in the code repository 122, or the like). The model management service 110 can store the updated model 116 and/or a model history 132 that can track two or more versions of the models 116. In some instances, metadata describing the models 116 and/or model histories 132 can be used to track the iterations of the model 116.

[0070] The model management service 110 can generate model data 128 (which can represent the model updates and model history events) and provide the updated model and model history 132 (or updated model history 132) to the

common repository **130** or other data storage resource. In various embodiments of the concepts and technologies disclosed herein, the model data **128** also can represent the model **116** in neutral format content (e.g., HTML content, XML, or the like) so that the model **116** can be viewed across platforms and/or by almost any user or device. It should be understood that this example embodiment is illustrative, and therefore should not be construed as being limiting in any way. Additional details of modifying and/or updating the models **116** will be illustrated and described in more detail below, particularly with reference to FIG. 4.

[0071] From operation **204**, the method **200** can proceed to operation **206**. At operation **206**, the server computer **112** can enable model and model history viewing. In some embodiments, the model management service **110** can receive a request to view a model **116** and/or a model history view (e.g., a view of a model history **132** with one or more links or embedded files representing iterations of the models **116** and/or related files such as diagrams, structures, code **120**, and the like). In response to detecting or receiving the request, the model management service **110** can be configured to identify the model **116** (associated with the requested model **116** and/or requested model history **132**) and obtain model data **128** associated with the identified model **116** and/or model history **132**. In various embodiments of the concepts and technologies disclosed herein, the model management service **110** can obtain the model data **128** from the common repository **130** and/or other data storage resources or structures.

[0072] The model management service **110** can be configured to generate, based on the model data **128**, a model history view, which can include renderable data that can be provided to one or more users, devices, or entities such as the user devices **102** for viewing and/or interacting with at the user devices **102**. The model history views can provide a graphical representation of the model history **132** and one or more models **116** represented in the model history **132**. Thus, embodiments of the concepts and technologies disclosed herein can enable viewing of an evolution of a model **116** during its life cycle from creation to a most recent iteration, and can enable access to associated models **116** and/or to related data such as diagrams, structures, or the like across the life cycle events. The model management service **110** can enable interactions with the model history view and provide updated iterations of the model history view at appropriate times in various embodiments. It should be understood that this example embodiment is illustrative, and therefore should not be construed as being limiting in any way. Additional details of model and model history viewing will be illustrated and described in more detail below, particularly with reference to FIG. 5.

[0073] From operation **206**, the method **200** can proceed to operation **208**. The method **200** can end at operation **208**.

[0074] Turning now to FIG. 3A, aspects of a method **300A** for creating and storing a model using a model management service will be described in detail, according to an illustrative embodiment. For purposes of illustrating and describing the concepts of the present disclosure, the method **300A** is described herein as being performed by the user device **102** via execution of one or more software modules such as, for example, the model management application **108**. It should be understood that additional and/or alternative devices and/or network nodes can provide the functionality described herein via execution of one or more modules,

applications, and/or other software including, but not limited to, the model management application **108**. Thus, the illustrated embodiments are illustrative, and should not be viewed as being limiting in any way.

[0075] The method **300A** begins at operation **302A**. At operation **302A**, the user device **102** can authenticate with a model management service **110**. Some embodiments of the model management service **110** can enforce an authentication requirement for users and/or devices attempting to create the models **116**. The model management service **110** can be configured to trigger authentication by other entities (e.g., an authentication server or module), or to include an authentication module or other functionality for providing these and/or other authentication services for the model management service **110**. Regardless of how the authentication is handled, operation **302A** can correspond to the user device **102** successfully authenticating to gain access to the model management service **110**. It should be understood that this example embodiment is illustrative, and therefore should not be construed as being limiting in any way.

[0076] From operation **302A**, the method **300A** can proceed to operation **304A**. At operation **304A**, the user device **102** can request and obtain code **120** from the model management service **110**. According to various embodiments of the concepts and technologies disclosed herein, an authenticated user or device such as the user device **102** can be configured to request, from the model management service **110**, code **120** associated with and/or representing a project, flow, process, product, network or component thereof, and/or the like, where this project is to be modeled. The model management service **110** can be configured to retrieve the code **120** (e.g., from a code repository **122**) and to provide the code **120** (e.g., as part of the model management data **114** and via a portal, API, webpage, or the like) to the user device **102** for analysis and/or viewing at the user device **102**. The user device **102** can be configured to store the code **120** in a local cache or other data storage device or resource. It should be understood that this example embodiment is illustrative, and therefore should not be construed as being limiting in any way.

[0077] From operation **304A**, the method **300A** can proceed to operation **306A**. At operation **306A**, the user device **102** can obtain and execute one or more scripts. According to various embodiments of the concepts and technologies disclosed herein, the user device **102** can obtain (e.g., as part of the model management data **114** from the server computer **112** or other entities) one or more scripts for execution at the user device **102**. The scripts can be executed by the user device **102A** (e.g., via execution of the model management application **108**) to create a model **116** based on the project, one or more templates **118**, and/or other data such as the code **120** and/or scripts. In some embodiments, the scripts can be executed to standardize (e.g., in accordance with defined templates **118**, guidelines, or the like) analysis of the model management data **114** to create the model **116** illustrated and described herein. In some embodiments, the scripts can be locally stored by the user device **102** (e.g., identifiable and/or assignable by the model management application **108**), though this is not the case in all embodiments. Regardless, it should be understood that during the creation phase, the user device **102** can be enabled to provide use of templates and scripts to enable “drag and drop” creation functions and/or the like, by standardizing representations of certain types of projects such as flows,

products, network devices, software, or the like. It should be understood that this example embodiment is illustrative, and therefore should not be construed as being limiting in any way.

[0078] From operation 306A, the method 300A can proceed to operation 308A. At operation 308A, the user device 102 can create a model 116 and/or related files such as directories, structures, and diagrams. The user device 102 can be configured to generate, based on the code 120, the scripts, one or more templates 118, and/or the project contours, one or more directories, one or more data structures, and/or one or more diagrams that collectively and/or individually can be considered the models 116 illustrated and described herein. It should be understood that this example embodiment is illustrative, and therefore should not be construed as being limiting in any way.

[0079] From operation 308A, the method 300A can proceed to operation 310A. At operation 310A, the user device 102 can load (or trigger the loading of) the model 116 to a common repository 130. In various embodiments of the concepts and technologies disclosed herein, the user device 102 can be configured to load the models 116 to a data storage resource (e.g., the common repository 130 in some embodiments), or to provide, to the server computer 112, modeling data to the server computer 112 and the server computer 112 can store the models 116 at the common repository 130 or elsewhere. It should be understood that this example embodiment is illustrative, and therefore should not be construed as being limiting in any way.

[0080] From operation 310A, the method 300A can proceed to operation 312A. The method 300A can end at operation 312A.

[0081] While FIG. 3A was described as being performed by the user device 102 via execution of the model management application 108, it should be understood that the functionality of FIG. 3A or substantially similar functionality can also be performed by the server computer 112 via execution of the model management service 110. As such, turning to FIG. 3B, aspects of a method 300B for creating and storing a model 116 using a model management service 110 will be described in detail, according to another illustrative embodiment. For purposes of illustrating and describing the concepts of the present disclosure, the method 300B is described herein as being performed by the server computer 112 via execution of one or more software modules such as, for example, the model management service 110. It should be understood that additional and/or alternative devices and/or network nodes can provide the functionality described herein via execution of one or more modules, applications, and/or other software including, but not limited to, the model management application 108. Thus, the illustrated embodiments are illustrative, and should not be viewed as being limiting in any way.

[0082] The method 300B begins and flow proceeds to operation 302B. At operation 302B the server computer 112 can authenticate the user device 102 (or detect connection of an authenticated user such as the user device 102). From operation 302B, the method 300B can proceed to operation 304B. In operation 304B, the server computer 112 can receive a request for the code 120 (for example from the user device 102) or other device, user, or entity, and provide the code 120 to the user device 102.

[0083] From operation 304B, the method 300B can proceed to operation 306B. In operation 306B, the server

computer 112 can trigger delivery of one or more scripts to the user device for execution. In some embodiments, the functionality of operation 306B can include the server computer 112 enabling access to scripts for the user device 102, though this is not necessarily the case in all embodiments.

[0084] From operation 306B, the method 300B can proceed to operation 308B. At operation 308B, the server computer 112 can obtain (e.g., from the user device 102) the model management data 114. In some embodiments, the server computer 112 can support creation of the model management data 114 via interactions with the user device 102 via a portal or API exposed by the server computer 112 instead of receiving the model management data 114.

[0085] From operation 308B, the method 300B can proceed to operation 310B. In operation 310B, the server computer 112 can create the model 116 and/or other structures related therewith such as directories, data structures, diagrams, and the like. From operation 310B, the method 300B can proceed to operation 312B. In operation 312B, the server computer 112 can load the model 116 to the common repository 130 or other data structure or resource. It can be appreciated that in loading the model 116 to the common repository, the server computer 112 can be configured to output the model 116 itself and/or to output model data 128 (e.g., neutral format content such as HTML, XML, or the like) that represents the model 116. It should be understood that this example embodiment is illustrative, and therefore should not be construed as being limiting in any way.

[0086] From operation 312B, the method 300B can proceed to operation 314B. The method 300B can end at operation 314B.

[0087] Turning now to FIG. 4, aspects of a method 400 for enriching, transforming, and updating a model using a model management service will be described in detail, according to an illustrative embodiment. For purposes of illustrating and describing the concepts of the present disclosure, the method 400 is described herein as being performed by the server computer 112 via execution of one or more software modules such as, for example, the model management service 110. It should be understood that additional and/or alternative devices and/or network nodes can provide the functionality described herein via execution of one or more modules, applications, and/or other software including, but not limited to, the model management service 110. Thus, the illustrated embodiments are illustrative, and should not be viewed as being limiting in any way.

[0088] The method 400 begins at operation 402. At operation 402, the server computer 112 can detect a model update event. According to various embodiments of the concepts and technologies disclosed herein, the model update event can include a transformation request, a job aid update, or the like.

[0089] In some embodiments, for example, a transformation operation can be performed in operation 402. For example, an event can be configured to extract one or more assets 124 from one or more asset catalogs 126 and the meta data associated with the one or more assets 124 can be transformed for inclusion in/modification of the model 116. This can be the case across platforms and/or languages and the model 116 can be updated accordingly.

[0090] In another contemplated embodiment, a job aid update by a system or actor can prompt the model management service 110 to update the model 116. The model

management service **110** can be configured to listen for events (e.g., changes to job aids), and upon detecting any such update, to prompt an update or enrichment in the model **116** (and optionally to create a history event for adding to the model history **132**) such as modeling one or more concepts in the job aid, updating one or more diagrams, or the like. It should be understood that this example embodiment is illustrative, and therefore should not be construed as being limiting in any way. Other types of update events can be detected by the model management service **110** as illustrated and described herein. As such, the above examples are illustrative and should not be construed as being limiting in any way.

[0091] From operation **402**, the method **400** can proceed to operation **404**. At operation **404**, the server computer **112** can update the model based on the detected update event. According to various embodiments of the concepts and technologies disclosed herein, the model management service **110** can be configured to obtain an asset **124** from an asset catalog **126**, and to incorporate the asset **124** (e.g., a representation of certain processes, systems, flows, or the like) into the model **116**. Thus, it can be appreciated that the assets **124** can represent portions of systems, flows, processes, methods, or the like; whereas the model **116** can represent, in some embodiments, the entire system, flow, process, method or the like (of which the asset **124** can represent a component or portion). It should be understood that this example embodiment is illustrative, and therefore should not be construed as being limiting in any way.

[0092] In some embodiments, the server computer **112** can also be configured to update the model **116** based on the job aid change. Thus, for example, if flow, process, system, device, or the like has been changed by the job aid change, the server computer **112** (e.g., via execution of the model management service **110**) can be configured to identify the changes to make those changes to the model **116** to reflect the updated job aid. It should be understood that this example embodiment is illustrative, and therefore should not be construed as being limiting in any way.

[0093] Other types of updates can be made to the models **116** in various embodiments of the concepts and technologies disclosed herein. As such, the above examples are illustrative of only some contemplated embodiments. As such, the above examples are illustrative and should not be construed as being limiting in any way.

[0094] From operation **404**, the method **400** can proceed to operation **406**. At operation **406**, the server computer **112** can update the model **116** and create an associated model history event. According to various embodiments of the concepts and technologies disclosed herein, the server computer **112** can be configured to obtain a past version of the model **116** and a new version of the model **116**. The server computer **112** can identify each component of each of the models **116** and compare the models **116**. Any differences between the models **116** can be tracked and a change event can be created, for example, a data flag that specifies that a change to the model **116** occurred at such-and-such a time by such-and-such a user or device. It should be understood that this example embodiment is illustrative, and therefore should not be construed as being limiting in any way.

[0095] From operation **406**, the method **400** can proceed to operation **408**. At operation **408**, the server computer **112** can store the updated model **116** and the updated model history **132**. Thus, the new model **116** can be stored with the

old model **116** and the events created by the server computer **112** can be stored as well. It should be understood that this example embodiment is illustrative, and therefore should not be construed as being limiting in any way.

[0096] From operation **408**, the method **400** can proceed to operation **410**. The method **400** can end at operation **410**.

[0097] Turning now to FIG. **5**, aspects of a method **500** for viewing and interacting with a model history using a model management service will be described in detail, according to an illustrative embodiment. For purposes of illustrating and describing the concepts of the present disclosure, the method **500** is described herein as being performed by the server computer **112** via execution of one or more software modules such as, for example, the model management service **110**. It should be understood that additional and/or alternative devices and/or network nodes can provide the functionality described herein via execution of one or more modules, applications, and/or other software including, but not limited to, the model management service **110**. Thus, the illustrated embodiments are illustrative, and should not be viewed as being limiting in any way.

[0098] The method **500** begins at operation **502**. At operation **502**, the server computer **112** can receive a request for a model history view (e.g., a model history **132** associated with a specified model **116**). According to various embodiments of the concepts and technologies disclosed herein, the server computer request received in operation **502** can correspond to a request from one or more of the user devices **102**, in some embodiments. In some embodiments, the server computer **112** can receive a request to view a model **116** and/or a model history view (e.g., a view of a model history **132** with one or more links or embedded files representing iterations of the models **116** and/or related files such as diagrams, structures, code **120**, and the like). It should be understood that this example embodiment is illustrative, and therefore should not be construed as being limiting in any way.

[0099] From operation **502**, the method **500** can proceed to operation **504**. At operation **504**, the server computer **112** can obtain model data **128** relating to the requested model history view (e.g., the model history **132** relating to the model **116** and/or relating to the model history **132** associated with the model **116**). In response to detecting or receiving the request, the server computer **112** can be configured to identify the model **116** (associated with the requested model **116** and/or requested model history **132**) and obtain model data **128** associated with the identified model **116** and/or model history **132**. In various embodiments of the concepts and technologies disclosed herein, the server computer **112** can obtain the model data **128** from the common repository **130** and/or other data storage resources or structures.

[0100] From operation **504**, the method **500** can proceed to operation **506**. At operation **506**, the server computer **112** can generate a model history view. The server computer **112** can be configured to generate, based on the model data **128**, a model history view. The model history view can include renderable data that can be provided to one or more users, devices, or entities such as the user devices **102** for viewing and/or interacting with at the user devices **102**. The model history views can provide a graphical representation of the model history **132** and one or more models **116** represented in the model history **132**. Thus, embodiments of the concepts and technologies disclosed herein can enable viewing of an

evolution of a model **116** during its life cycle from creation to a most recent iteration, and can enable access to associated models **116** and/or to related data such as diagrams, structures, or the like across the life cycle events. It should be understood that this example embodiment is illustrative, and therefore should not be construed as being limiting in any way.

[0101] From operation **506**, the method **500** can proceed to operation **508**. At operation **508**, the server computer **112** can enable interactions with the model history view. The server computer **112** can enable interactions with the model history view and provide updated iterations of the model history view at appropriate times in various embodiments. Operation **508** can include the server computer **112** exposing the model history view to the user or other entity (e.g., the user device **102**) to enable modifications and/or interactions such as viewing model **116** and/or model components, modifying models **116** and/or model components, or the like. Changes can be saved and included in the model histories **132**. It should be understood that this example embodiment is illustrative, and therefore should not be construed as being limiting in any way.

[0102] From operation **508**, the method **500** can proceed to operation **510**. The method **500** can end at operation **510**.

[0103] FIGS. 6A-6C are user interface (“UI”) diagrams showing aspects of some example UIs for using and/or interacting with a model management service **110** and/or the model management application **108**, according to some illustrative embodiments. FIG. 6A shows an illustrative screen display **600A**. According to some embodiments of the concepts and technologies described herein, the screen display **600A** can be generated by a device such as the user device **102** via interactions with the model management service **110** and/or the model management application **108**. In particular, according to various embodiments, the user device **102** can generate the screen display **600A** and/or other screen displays in conjunction with and/or based upon interactions with the model management application **108** described herein, which can be configured to render the screen display **600A** by rendering data generated at the user device **102** and/or using data provided by the model management service **110** (e.g., the model management data **114**). It should be appreciated that the UI diagram illustrated in FIG. 6A is illustrative of one contemplated example of the UIs that can be generated and/or displayed in accordance with the concepts and technologies disclosed herein, and therefore should not be construed as being limited in any way.

[0104] According to various embodiments, the screen display **600A** can be presented, for example, when a user or other entity requests a model history and/or model view as illustrated and described herein. Because the screen display **600A** illustrated in FIG. 6A can be displayed at additional and/or alternative times, it should be understood that these examples are illustrative and therefore should not be construed as being limiting in any way.

[0105] The screen display **600A** can include various menus and/or menu options (not shown in FIG. 6A). The screen display **600A** also can include a model view window **602**. The model view window **602** can be configured to present one or more components of a model **116** such as, for example, a diagram (as shown in FIG. 6A), or the like. The model view window **602** can include various types of data and/or information that can be included such as, for

example, a model history indicator **604** that can provide information regarding the model creation date, model modification data, and/or other history events as illustrated and described herein with reference to the model histories **132**. Thus, as illustrated and described herein, the model view window **602** can provide a view of the model **116** and the model history **132** (or a portion thereof) in some embodiments. It should be understood that this example embodiment is illustrative, and therefore should not be construed as being limiting in any way.

[0106] The model view window **602** also can include a model navigation menu **606** that relates to the model **116**. As shown in the example embodiment illustrated in FIG. 6A, the model navigation menu **606** can provide links to other components of the model **116** such as, for example, definitions associated with the model **116** and/or the model and/or model history display, other version of the model **116** and/or model components, etc. It should be understood that the illustrated menu **606** is illustrative, and that the menu **606** can include additional and/or alternative information as illustrated and described herein. It should be understood that this example is illustrative and therefore should not be construed as being limiting in any way. Because additional or alternative controls can be included in the model view window **602**, it should be understood that the example embodiment shown in FIG. 6A is illustrative and therefore should not be construed as being limiting in any way.

[0107] FIG. 6B shows an illustrative screen display **600B**. According to some embodiments of the concepts and technologies described herein, the screen display **600B** can be generated by a device such as the user device **102** via interactions with the model management service **110** and/or the model management application **108**. In particular, according to various embodiments, the user device **102** can generate the screen display **600B** and/or other screen displays in conjunction with and/or based upon interactions with the model management application **108** described herein, which can be configured to render the screen display **600B** by rendering data generated at the user device **102** and/or using data provided by the model management service **110** (e.g., the model management data **114**). It should be appreciated that the UI diagram illustrated in FIG. 6B is illustrative of one contemplated example of the UIs that can be generated and/or displayed in accordance with the concepts and technologies disclosed herein, and therefore should not be construed as being limited in any way.

[0108] According to various embodiments, the screen display **600B** can be presented, for example, when a user or other entity selects an option to view definitions from the menu **606** shown in the screen display **600A** of FIG. 6A. Because the screen display **600B** illustrated in FIG. 6B can be displayed at additional and/or alternative times, it should be understood that these examples are illustrative and therefore should not be construed as being limiting in any way.

[0109] The screen display **600B** can include various menus and/or menu options (not shown in FIG. 6B). The screen display **600B** also can include a model explanation view **608**. The model explanation view **608** can be configured to present one or more documents and/or links explaining various aspects of the model **116** such as, for example, documents with definitions, process and/or system flows, and the like. The model explanation view **608** can include various types of data and/or information that can be included such as, for example, documents, links to web pages, links

to online resources, and the like. Thus, as illustrated and described herein, the model explanation view **608** can provide almost any sort of information that can explain the model **116** and the model history **132** (or a portion thereof) in some embodiments. It should be understood that this example embodiment is illustrative, and therefore should not be construed as being limiting in any way. Because additional or alternative controls can be included in the model explanation view **608**, it should be understood that the example embodiment shown in FIG. **6B** is illustrative and therefore should not be construed as being limiting in any way.

[0110] FIG. **6C** shows an illustrative screen display **600C**. According to some embodiments of the concepts and technologies described herein, the screen display **600C** can be generated by a device such as the user device **102** via interactions with the model management service **110** and/or the model management application **108**. In particular, according to various embodiments, the user device **102** can generate the screen display **600C** and/or other screen displays in conjunction with and/or based upon interactions with the model management application **108** described herein, which can be configured to render the screen display **600C** by rendering data generated at the user device **102** and/or using data provided by the model management service **110** (e.g., the model management data **114**). It should be appreciated that the UI diagram illustrated in FIG. **6C** is illustrative of one contemplated example of the UIs that can be generated and/or displayed in accordance with the concepts and technologies disclosed herein, and therefore should not be construed as being limited in any way.

[0111] According to various embodiments, the screen display **600C** can be presented, for example, when a user or other entity selects an option to view a model history **132** associated with the model **116**. Because the screen display **600C** illustrated in FIG. **6C** can be displayed at additional and/or alternative times, it should be understood that these examples are illustrative and therefore should not be construed as being limiting in any way.

[0112] The screen display **600C** can include various menus and/or menu options (not shown in FIG. **6C**). The screen display **600C** also can include a model history details view **610**. The model history details view **610** can be configured to present one or more aspects of a model history **132** such as a list of events **612**, various menu controls **614**, and another section **616** of relevant information relating to the model **116** and/or the model history **132**. The model history details view **610** can include various types of data and/or information that can be included such as, for example, documents, links to web pages, links to online resources, and the like, as well as detailed information relating to the history of the model **116** and relevant information relating thereto. Thus, as illustrated and described herein, the model history details view **610** can provide almost any sort of information that can explain the model **116** and the model history **132** (or a portion thereof) in some embodiments. It should be understood that this example embodiment is illustrative, and therefore should not be construed as being limiting in any way. Because additional or alternative controls can be included in the model history details view **610**, it should be understood that the example embodiment shown in FIG. **6C** is illustrative and therefore should not be construed as being limiting in any way.

[0113] Turning now to FIGS. **7A-7H**, some example call flow diagrams illustrating additional aspects of the concepts and technologies disclosed herein for creating, storing modifying, and interacting with a model using a model management service will be described. FIG. **7A** illustrates an example call flow diagram showing a process **700A** for preprocessing access and artifacts for a model **116**. This example process **700A** can be performed to enable an isolated a system or actor (e.g., the user device **102A** alone and/or in cooperation with a model management service **110**) that is responsible for modeling to share created models at some point in the future. Furthermore, the process **700A** can enable an acquisition of one or more template models (e.g., the templates **118** shown in FIG. **1**) in many different formats. In some embodiments, almost any format can be used where the format is a common extension to the base modeling with common or shared end-user decorations of models for publication in neutral job aids by commonly understood user-defined properties. It can be appreciated with reference to FIG. **7A** that having common template models (e.g., the templates **118**) with common extensions to the base modeling can enable people without advanced modeling tool knowledge to engineer and update the models **116**. Furthermore, such an approach can enable systems and/or AI to absorb well-defined contexts and common processing pipelines. It should be understood that this example embodiment is illustrative, and therefore should not be construed as being limiting in any way.

[0114] FIG. **7B** illustrates an example call flow diagram showing a process **700B** for preprocessing model extraction. This example process **700B** can be performed to enable a system or actor (e.g., the user device **102A** alone and/or in cooperation with a model management service **110**) to save a local model and extra neutral format content to be shared for external viewing (e.g., HTML reports, XML files, etc.) and for additional processing and/or automated processing. Furthermore, the process **700B** can be performed in an environment that can support neutral content (e.g., XML or XMI) for common back-end processing of content in standards such as Open Data Standard, Open API, Open AI, or the like, via automation functions and/or providing the output to a cloud repository such as the common repository **130**. Such processes can be enabled by way of drive mapping without concern for the vendor-specific model binary format for tools such as ERWIN (i.e., the data modeling platform known as “ERWIN”), Sparx EA (e.g., the Enterprise Architect and/or other families of modeling and architecture tools from SPARX), or the like. It should be understood that this example embodiment is illustrative, and therefore should not be construed as being limiting in any way.

[0115] FIG. **7C** illustrates an example call flow diagram showing a process **700C** for enriching a model **116**. This example process **700C** can be performed to enable a system or actor (e.g., the user device **102A** alone and/or in cooperation with a model management service **110**) to update a model **116** (e.g., a model **116** stored in the common repository **130**) by updating a job aid. For example, a spreadsheet (e.g., a MICROSOFT EXCEL spreadsheet or GOOGLE SHEETS spreadsheet) that interacts with events and triggers the repository to update concepts, contexts, or other content in the cloud and/or cloud model repository (e.g., the common repository) can trigger a model enrichment process when updated. FIG. **7C** illustrates an example process **700C**

for such an update. It should be understood that this example embodiment is illustrative, and therefore should not be construed as being limiting in any way.

[0116] FIG. 7D illustrates an example call flow diagram showing a process 700D for multi-system rationalization (e.g., propagating updates to the model 116 and/or other systems or catalogs such as asset catalog 126, the code repository 122, and/or the common repository 130). This example process 700D can be performed to enable a system or actor (e.g., the user device 102A alone and/or in cooperation with a model management service 110) to update code and model repositories (e.g., the code repository 122 and the common repository 130) to update any number of models 116 and/or related files (e.g., assets 124 in the asset catalog 126, code 120 in the code repository 122, models 116, model data 128, and/or model histories 132 in the common repository 130, or the like). Furthermore, the process 700D can trigger updates to events (e.g., model history events and/or model histories 132), the templates 118, and/or other data catalogs, feature stores, event catalogs, and/or API/service catalogs. It should be understood that this example embodiment is illustrative, and therefore should not be construed as being limiting in any way.

[0117] FIG. 7E illustrates an example call flow diagram showing a process 700E for performing target transformation for a model 116. This example process 700E can be performed to enable a system or actor (e.g., the user device 102A alone and/or in cooperation with a model management service 110) to trigger an event to extra well-known named assets 124 in the asset catalog 126 (or multiple asset catalogs 126) and to promote inter-model communication with local and remote caches. In one example implementation of the process 700E, a chosen business-ready dataset (BRD) can be extracted from a cloud-based model repository (e.g., the common repository 130), the code repository 122, and the one or more asset catalogs 126. This extraction can be performed regardless of the models 116 being in different formats (e.g., modeling tool formats and/or binary formats such as ERWIN, Sparx EA, or the like). It should be understood that this example embodiment is illustrative, and therefore should not be construed as being limiting in any way.

[0118] FIG. 7F illustrates an example call flow diagram showing a process 700F for updating various catalogs and/or APIs (e.g., a distributed synchronization). This example process 700F can be performed to generate by a system or actor (e.g., the user device 102A alone and/or in cooperation with a model management service 110) an event to indicate a change or update to a model 116 or a file associated therewith. The model management service 110 can be configured to open CSV results (e.g., if generated at the end of the process 700E illustrated herein), create standard format update based on the CSV results (e.g., a JSON file or the like), and then authenticate with and update (e.g., via RESTful API calls with the asset catalog 126) the asset catalog 126. It should be understood that this example embodiment is illustrative, and therefore should not be construed as being limiting in any way.

[0119] FIGS. 7G-7H illustrate example processes 700G and 700H for post-processing operations that can be performed in accordance with the concepts and technologies disclosed herein. The processes 700G and 700H can be triggered by a system or actor (e.g., the user device 102A alone and/or in cooperation with a model management

service 110) or other entities. It should be understood that this example embodiment is illustrative, and therefore should not be construed as being limiting in any way.

[0120] Turning now to FIG. 8, additional details of the network 104 are illustrated, according to an illustrative embodiment. The network 104 includes a cellular network 802, a packet data network 804, for example, the Internet, and a circuit switched network 806, for example, a publicly switched telephone network ("PSTN"). The cellular network 802 includes various components such as, but not limited to, base transceiver stations ("BTSS"), NodeB's or eNodeB's ("eNBs"), gNodeBs ("gNBs"), or the like; base station controllers ("BSCs") radio network controllers ("RNCs"), or the like; an evolved packet core ("EPC"); mobile switching centers ("MSCs" or "MSSs"); session management functions ("SMFs"); mobile management entities ("MMEs"); access and mobility management functions ("AMFs"); authentication server functions ("AUSFs"), network slice selection functions ("NSSFs"); network exposure functions ("NEFs"); policy control functions ("PCFs"); and various other functions in the user and control planes such as, for example, user plane functions ("UPFs"), application functions ("AFs"), NF repository functions ("NRFs"), and the like; short message service centers ("SMSCs"); multimedia messaging service centers ("MMSCs"); home location registers ("HLRs"); home subscriber servers ("HSSs"); visitor location registers ("VLRs"); charging platforms; billing platforms; voicemail platforms; GPRS core network components; links to data networks ("DNs") and/or other operator services, third party services, and/or the Internet; location service nodes, an IP Multimedia Subsystem ("IMS"); and the like. Of course, the cellular network 802 also can include various interfaces between various components, as is generally understood. The cellular network 802 also includes radios and nodes for receiving and transmitting voice, data, and combinations thereof to and from radio transceivers, networks, the packet data network 804, and the circuit switched network 806.

[0121] A mobile communications device 808, such as, for example, a cellular telephone, a user equipment, a mobile terminal, a PDA, a laptop computer, a handheld computer, and combinations thereof, can be operatively connected to the cellular network 802. The cellular network 802 can be configured as a 2G GSM network and can provide data communications via GPRS and/or EDGE. Additionally, or alternatively, the cellular network 802 can be configured as a 3G UMTS network and can provide data communications via the HSPA protocol family, for example, HSDPA, EUL (also referred to as HSUPA), and HSPA+. The cellular network 802 also is compatible with 4G mobile communications standards, 5G mobile communications standards, 6G mobile communication standards, other mobile communications standards, and evolved and future mobile communications standards.

[0122] The packet data network 804 includes various devices, for example, servers, computers, databases, and other devices in communication with one another, as is generally known. The packet data network 804 devices are accessible via one or more network links. The servers often store various files that are provided to a requesting device such as, for example, a computer, a terminal, a smartphone, or the like. Typically, the requesting device includes software (a "browser") for executing a web page in a format readable by the browser or other software. Other files and/or

data may be accessible via “links” in the retrieved files, as is generally known. In some embodiments, the packet data network **804** includes or is in communication with the Internet. The circuit switched network **806** includes various hardware and software for providing circuit switched communications. The circuit switched network **806** may include, or may be, what is often referred to as a plain old telephone system (POTS). The functionality of a circuit switched network **806** or other circuit-switched network are generally known and will not be described herein in detail.

[0123] The illustrated cellular network **802** is shown in communication with the packet data network **804** and a circuit switched network **806**, though it should be appreciated that this is not necessarily the case. One or more Internet-capable devices **810**, for example, a PC, a laptop, a portable device, or another suitable device, can communicate with one or more cellular networks **802**, and devices connected thereto, through the packet data network **804**. It also should be appreciated that the Internet-capable device **810** can communicate with the packet data network **804** through the circuit switched network **806**, the cellular network **802**, and/or via other networks (not illustrated).

[0124] As illustrated, a communications device **812**, for example, a telephone, facsimile machine, modem, computer, or the like, can be in communication with the circuit switched network **806**, and therethrough to the packet data network **804** and/or the cellular network **802**. It should be appreciated that the communications device **812** can be an Internet-capable device, and can be substantially similar to the Internet-capable device **810**. In the specification, the network **104** is used to refer broadly to any combination of the networks **802**, **804**, **806**. It should be appreciated that substantially all of the functionality described with reference to the network **104** can be performed by the cellular network **802**, the packet data network **804**, and/or the circuit switched network **806**, alone or in combination with other networks, network elements, and the like.

[0125] FIG. 9 is a block diagram illustrating a computer system **900** configured to provide the functionality described herein for providing model creation and management using a model management service, in accordance with various embodiments of the concepts and technologies disclosed herein. As such, it can be appreciated that the user devices **102** and/or the server computer **112** can have an architecture similar to the computer system **900** shown in FIG. 9 in some embodiments, though this is not necessarily the case. The computer system **900** includes a processing unit **902**, a memory **904**, one or more user interface devices **906**, one or more input/output (“I/O”) devices **908**, and one or more network devices **910**, each of which is operatively connected to a system bus **912**. The system bus **912** can enable bi-directional communication between the processing unit **902**, the memory **904**, the user interface devices **906**, the I/O devices **908**, and the network devices **910**.

[0126] The processing unit **902** may be a standard central processor that performs arithmetic and logical operations, a more specific purpose programmable logic controller (“PLC”), a programmable gate array, or other type of processor known to those skilled in the art and suitable for controlling the operation of the server computer. As used herein, the word “processor” and/or the phrase “processing unit” when used with regard to any architecture or system can include multiple processors or processing units distributed across and/or operating in parallel in a single machine

or in multiple machines. Furthermore, processors and/or processing units can be used to support virtual processing environments. Processors and processing units also can include state machines, application-specific integrated circuits (“ASICs”), combinations thereof, or the like. Because processors and/or processing units are generally known, the processors and processing units disclosed herein will not be described in further detail herein.

[0127] The memory **904** communicates with the processing unit **902** via the system bus **912**. In some embodiments, the memory **904** is operatively connected to a memory controller (not shown) that enables communication with the processing unit **902** via the system bus **912**. The memory **904** includes an operating system **914** and one or more program modules **916**. The operating system **914** can include, but is not limited to, members of the WINDOWS, WINDOWS CE, and/or WINDOWS MOBILE families of operating systems from MICROSOFT CORPORATION, the LINUX family of operating systems, the SYMBIAN family of operating systems from SYMBIAN LIMITED, the BREW family of operating systems from QUALCOMM CORPORATION, the MAC OS, iOS, and/or SONOMA families of operating systems from APPLE CORPORATION, the FREEBSD family of operating systems, the SOLARIS family of operating systems from ORACLE CORPORATION, other operating systems, and the like.

[0128] The program modules **916** may include various software and/or program modules described herein. In some embodiments, for example, the program modules **916** include the model management application **108** and/or the server computer **112**. This and/or other programs can be embodied in computer-readable media containing instructions that, when executed by the processing unit **902**, perform one or more of the methods **200**, **300**, **400**, **500** described in detail above with respect to FIGS. 2-5 and/or other functionality as illustrated and described herein. It can be appreciated that, at least by virtue of the instructions embodying the methods **200**, **300**, **400**, **500**, and/or other functionality illustrated and described herein being stored in the memory **904** and/or accessed and/or executed by the processing unit **902**, the computer system **900** is a special-purpose computing system that can facilitate providing the functionality illustrated and described herein. According to embodiments, the program modules **916** may be embodied in hardware, software, firmware, or any combination thereof. Although not shown in FIG. 9, it should be understood that the memory **904** also can be configured to store the model management data **114**, the models **116**, the templates **118**, the code **120**, the assets **124**, the model data **128**, the model histories **132**, the metadata **134**, and/or other data, if desired.

[0129] By way of example, and not limitation, computer-readable media may include any available computer storage media or communication media that can be accessed by the computer system **900**. Communication media includes computer-readable instructions, data structures, program modules, or other data in a modulated data signal such as a carrier wave or other transport mechanism and includes any delivery media. The term “modulated data signal” means a signal that has one or more of its characteristics changed or set in a manner as to encode information in the signal. By way of example, and not limitation, communication media includes wired media such as a wired network or direct-wired connection, and wireless media such as acoustic, RF,

infrared and other wireless media. Combinations of any of the above should also be included within the scope of computer-readable media.

[0130] Computer storage media includes only non-transitory embodiments of computer readable media as illustrated and described herein. Thus, computer storage media can include volatile and non-volatile, removable and non-removable media implemented in any method or technology for storage of information such as computer-readable instructions, data structures, program modules, or other data. Computer storage media includes, but is not limited to, RAM, ROM, Erasable Programmable ROM (“EPROM”), Electrically Erasable Programmable ROM (“EEPROM”), flash memory or other solid state memory technology, CD-ROM, digital versatile disks (“DVD”), or other optical storage, magnetic cassettes, magnetic tape, magnetic disk storage or other magnetic storage devices, or any other medium which can be used to store the desired information and which can be accessed by the computer system 900. In the claims, the phrase “computer storage medium” and variations thereof does not include waves or signals per se and/or communication media.

[0131] The user interface devices 906 may include one or more devices with which a user accesses the computer system 900. The user interface devices 906 may include, but are not limited to, computers, servers, personal digital assistants, cellular phones, or any suitable computing devices. The I/O devices 908 enable a user to interface with the program modules 916. In one embodiment, the I/O devices 908 are operatively connected to an I/O controller (not shown) that enables communication with the processing unit 902 via the system bus 912. The I/O devices 908 may include one or more input devices, such as, but not limited to, a keyboard, a mouse, or an electronic stylus. Further, the I/O devices 908 may include one or more output devices, such as, but not limited to, a display screen or a printer.

[0132] The network devices 910 enable the computer system 900 to communicate with other networks or remote systems via a network, such as the network 104. Examples of the network devices 910 include, but are not limited to, a modem, a radio frequency (“RF”) or infrared (“IR”) transceiver, a telephonic interface, a bridge, a router, or a network card. The network 104 may include a wireless network such as, but not limited to, a Wireless Local Area Network (“WLAN”) such as a Wi-Fi network, a Wireless Wide Area Network (“WWAN”), a Wireless Personal Area Network (“WPAN”) such as BLUETOOTH, a Wireless Metropolitan Area Network (“WMAN”) such as a WiMAX network, or a cellular network. Alternatively, the network 104 may be a wired network such as, but not limited to, a Wide Area Network (“WAN”) such as the Internet, a Local Area Network (“LAN”) such as the Ethernet, a wired Personal Area Network (“PAN”), or a wired Metropolitan Area Network (“MAN”).

[0133] FIG. 10 illustrates an illustrative architecture for a cloud computing platform 1000 that can be capable of executing the software components described herein for providing model creation and management using a model management service 110 and/or for interacting with the model management application 108 and/or the model management service 110. Thus, it can be appreciated that in some embodiments of the concepts and technologies disclosed herein, the cloud computing platform 1000 illustrated in FIG. 10 can be used to provide the functionality described

herein with respect to the user devices 102, the server computer 112, the code repository 122, the asset catalog 126, the common repository 130, and/or the metadata sources 136.

[0134] The cloud computing platform 1000 thus may be utilized to execute any aspects of the software components presented herein. Thus, according to various embodiments of the concepts and technologies disclosed herein, the model management application 108 and/or the model management service 110 can be implemented, at least in part, on or by elements included in the cloud computing platform 1000 illustrated and described herein. Those skilled in the art will appreciate that the illustrated cloud computing platform 1000 is a simplification of but only one possible implementation of an illustrative cloud computing platform, and as such, the illustrated cloud computing platform 1000 should not be construed as being limiting in any way.

[0135] In the illustrated embodiment, the cloud computing platform 1000 can include a hardware resource layer 1002, a virtualization/control layer 1004, and a virtual resource layer 1006. These layers and/or other layers can be configured to cooperate with each other and/or other elements of a cloud computing platform 1000 to perform operations as will be described in detail herein. While connections are shown between some of the components illustrated in FIG. 10, it should be understood that some, none, or all of the components illustrated in FIG. 10 can be configured to interact with one another to carry out various functions described herein. In some embodiments, the components are arranged so as to communicate via one or more networks such as, for example, the network 104 illustrated and described hereinabove (not shown in FIG. 10). Thus, it should be understood that FIG. 10 and the following description are intended to provide a general understanding of a suitable environment in which various aspects of embodiments can be implemented, and should not be construed as being limiting in any way.

[0136] The hardware resource layer 1002 can provide hardware resources. In the illustrated embodiment, the hardware resources can include one or more compute resources 1008, one or more memory resources 1010, and one or more other resources 1012. The compute resource(s) 1008 can include one or more hardware components that can perform computations to process data, and/or to execute computer-executable instructions of one or more application programs, operating systems, services, and/or other software including, but not limited to, the operating system 106, the model management application 108, and/or the model management service 110 illustrated and described herein.

[0137] According to various embodiments, the compute resources 1008 can include one or more central processing units (“CPUs”). The CPUs can be configured with one or more processing cores. In some embodiments, the compute resources 1008 can include one or more graphics processing units (“GPUs”). The GPUs can be configured to accelerate operations performed by one or more CPUs, and/or to perform computations to process data, and/or to execute computer-executable instructions of one or more application programs, operating systems, and/or other software that may or may not include instructions that are specifically graphics computations and/or related to graphics computations. In some embodiments, the compute resources 1008 can include one or more discrete GPUs. In some other embodiments, the compute resources 1008 can include one or more CPU

and/or GPU components that can be configured in accordance with a co-processing CPU/GPU computing model. Thus, it can be appreciated that in some embodiments of the compute resources **1008**, a sequential part of an application can execute on a CPU and a computationally-intensive part of the application can be accelerated by the GPU. It should be understood that this example is illustrative, and therefore should not be construed as being limiting in any way.

[0138] In some embodiments, the compute resources **1008** also can include one or more system on a chip (“SoC”) component. It should be understood that an SoC component can operate in association with one or more other components as illustrated and described herein, for example, one or more of the memory resources **1010** and/or one or more of the other resources **1012**. In some embodiments in which an SoC component is included, the compute resources **1008** can be or can include one or more embodiments of the SNAP-DRAGON brand family of SoCs, available from QUALCOMM of San Diego, California; one or more embodiment of the TEGRA brand family of SoCs, available from NVIDIA of Santa Clara, California; one or more embodiment of the HUMMINGBIRD brand family of SoCs, available from SAMSUNG of Seoul, South Korea; one or more embodiment of the Open Multimedia Application Platform (“OMAP”) family of SoCs, available from TEXAS INSTRUMENTS of Dallas, Texas; one or more customized versions of any of the above SoCs; and/or one or more other brand and/or one or more proprietary SoCs.

[0139] The compute resources **1008** can be or can include one or more hardware components arranged in accordance with an ARM architecture, available for license from ARM HOLDINGS of Cambridge, United Kingdom. Alternatively, the compute resources **1008** can be or can include one or more hardware components arranged in accordance with an x86 architecture, such as an architecture available from INTEL CORPORATION of Mountain View, California, and others. Those skilled in the art will appreciate the implementation of the compute resources **1008** can utilize various computation architectures and/or processing architectures. As such, the various example embodiments of the compute resources **1008** as mentioned hereinabove should not be construed as being limiting in any way. Rather, implementations of embodiments of the concepts and technologies disclosed herein can be implemented using compute resources **1008** having any of the particular computation architecture and/or combination of computation architectures mentioned herein as well as other architectures.

[0140] Although not separately illustrated in FIG. 10, it should be understood that the compute resources **1008** illustrated and described herein can host and/or execute various services, applications, portals, and/or other functionality illustrated and described herein. Thus, the compute resources **1008** can host and/or can execute the model management application **108**, the model management service **110**, and/or other applications or services illustrated and described herein.

[0141] The memory resource(s) **1010** can include one or more hardware components that can perform or provide storage operations, including temporary and/or permanent storage operations. In some embodiments, the memory resource(s) **1010** can include volatile and/or non-volatile memory implemented in any method or technology for storage of information such as computer-readable instructions, data structures, program modules, or other data dis-

closed herein. Computer storage media is defined hereinabove and therefore should be understood as including, in various embodiments, random access memory (“RAM”), read-only memory (“ROM”), Erasable Programmable ROM (“EPROM”), Electrically Erasable Programmable ROM (“EEPROM”), flash memory or other solid state memory technology, CD-ROM, digital versatile disks (“DVD”), or other optical storage, magnetic cassettes, magnetic tape, magnetic disk storage or other magnetic storage devices, or any other medium that can be used to store data and that can be accessed by the compute resources **1008**, subject to the definition of “computer storage media” provided above (e.g., as excluding waves and signals per se and/or communication media as defined in this application).

[0142] Although not illustrated in FIG. 10, it should be understood that the memory resources **1010** can host or store the various data illustrated and described herein including, but not limited to, the model management data **114**, the models **116**, the templates **118**, the code **120**, the assets **124**, the model data **128**, the model histories **132**, the metadata **134**, and/or other data, if desired. It should be understood that this example is illustrative, and therefore should not be construed as being limiting in any way.

[0143] The other resource(s) **1012** can include any other hardware resources that can be utilized by the compute resources(s) **1008** and/or the memory resource(s) **1010** to perform operations. The other resource(s) **1012** can include one or more input and/or output processors (e.g., a network interface controller and/or a wireless radio), one or more modems, one or more codec chipsets, one or more pipeline processors, one or more fast Fourier transform (“FFT”) processors, one or more digital signal processors (“DSPs”), one or more speech synthesizers, combinations thereof, or the like.

[0144] The hardware resources operating within the hardware resource layer **1002** can be virtualized by one or more virtual machine monitors (“VMMs”) **1014A-1014N** (also known as “hypervisors;” hereinafter “VMMs **1014**”). The VMMs **1014** can operate within the virtualization/control layer **1004** to manage one or more virtual resources that can reside in the virtual resource layer **1006**. The VMMs **1014** can be or can include software, firmware, and/or hardware that alone or in combination with other software, firmware, and/or hardware, can manage one or more virtual resources operating within the virtual resource layer **1006**.

[0145] The virtual resources operating within the virtual resource layer **1006** can include abstractions of at least a portion of the compute resources **1008**, the memory resources **1010**, the other resources **1012**, or any combination thereof. These abstractions are referred to herein as virtual machines (“VMs”). In the illustrated embodiment, the virtual resource layer **1006** includes VMs **1016A-1016N** (hereinafter “VMs **1016**”).

[0146] Based on the foregoing, it should be appreciated that systems and methods for providing model creation and management using a model management service have been disclosed herein. Although the subject matter presented herein has been described in language specific to computer structural features, methodological and transformative acts, specific computing machinery, and computer-readable media, it is to be understood that the concepts and technologies disclosed herein are not necessarily limited to the specific features, acts, or media described herein. Rather, the

specific features, acts and mediums are disclosed as example forms of implementing the concepts and technologies disclosed herein.

[0147] The subject matter described above is provided by way of illustration only and should not be construed as limiting. Various modifications and changes may be made to the subject matter described herein without following the example embodiments and applications illustrated and described, and without departing from the true spirit and scope of the embodiments of the concepts and technologies disclosed herein.

1. A system comprising:
 - a processor; and
 - a memory that stores computer-executable instructions that, when executed by the processor, cause the processor to perform operations comprising
 - obtaining, from an authenticated user device, model management data that comprises modeling data that defines a project to model, wherein the project comprises a network flow, a network system, a device, or a process,
 - detecting, in the model management data, a selection of a template to apply to the project to generate the model,
 - creating, based on the model management data and the template, the model, wherein the model comprises a directory, a data structure, and a diagram that describes the project,
 - converting the model into a device-agnostic format version of the model, and
 - loading, to a common repository, model data that comprises the device-agnostic format version of the model.
2. The system of claim 1, wherein the device-agnostic format version of the model comprises hypertext markup language or extensible markup language.
3. The system of claim 1, wherein the common repository also stores a model history associated with the model, wherein the model history comprises data that describes creation of the model.
4. The system of claim 1, wherein the computer-executable instructions, when executed by the processor, cause the processor to perform operations further comprising:
 - detecting a model update comprising a request to update the model;
 - retrieving, from the common repository, the model;
 - updating the model based on the model update;
 - generating a model update event that defines an update to the model;
 - adding the model update event to a model history stored at the common repository; and
 - storing an updated model and an updated model history at the common repository.
5. The system of claim 4, wherein the model update event comprises an addition to the model, and wherein updating the model comprises:
 - obtaining an asset from an asset catalog, wherein the asset relates to the addition; and
 - updating the model based on the asset and the model update event.

6. The system of claim 1, wherein the computer-executable instructions, when executed by the processor, cause the processor to perform operations further comprising:

- receiving a request for a model history view relating to the model;
- obtaining model data and a model history from the common repository, wherein the model data describes the model and wherein the model history describes lifecycle events associated with the model;
- generating the model history view, wherein the model history view depicts model update events associated with the model and versions of the model; and
- enabling interactions with the model history view by the authenticated user device via portal.

7. The system of claim 6, wherein the model history view comprises depictions of at least two model update events associated with the model, the model update events being obtained from the model history, and wherein the model history view provides access to at least two versions of the model.

8. A method comprising:

- obtaining, at a server computer comprising a processor and from an authenticated user device, model management data that comprises modeling data that defines a project to model, wherein the project comprises a network flow, a network system, a device, or a process;
- detecting, by the processor and in the model management data, a selection of a template to apply to the project to generate the model;
- creating, by the processor and based on the model management data and the template, the model, wherein the model comprises a directory, a data structure, and a diagram that describes the project;
- converting, by the processor, the model into a device-agnostic format version of the model; and
- loading, by the processor and to a common repository, model data that comprises the device-agnostic format version of the model.

9. The method of claim 8, wherein the device-agnostic format version of the model comprises hypertext markup language or extensible markup language.

10. The method of claim 8, wherein the common repository also stores a model history associated with the model, wherein the model history comprises data that describes creation of the model.

11. The method of claim 8, further comprising:

- detecting, by the processor, a model update comprising a request to update the model;
- retrieving, by the processor and from the common repository, the model;
- updating, by the processor, the model based on the model update;
- generating, by the processor, a model update event that defines an update to the model;
- adding, by the processor, the model update event to a model history stored at the common repository; and
- storing, by the processor, an updated model and an updated model history at the common repository.

12. The method of claim 11, wherein the model update event comprises an addition to the model, and wherein updating the model comprises:

- obtaining an asset from an asset catalog, wherein the asset relates to the addition; and
- updating the model based on the asset and the model update event.

13. The method of claim 8, further comprising:

- receiving a request for a model history view relating to the model;

- obtaining model data and a model history from the common repository, wherein the model data describes the model and wherein the model history describes lifecycle events associated with the model;

- generating the model history view, wherein the model history view comprises depictions of at least two model update events associated with the model, the at least two model update events being obtained from the model history, and wherein the model history view provides access to at least two versions of the model; and

- enabling interactions with the model history view by the authenticated user device via portal.

14. A computer storage medium having computer-executable instructions stored thereon that, when executed by a processor, cause the processor to perform operations comprising:

- obtaining, from an authenticated user device, model management data that comprises modeling data that defines a project to model, wherein the project comprises a network flow, a network system, a device, or a process;
- detecting, in the model management data, a selection of a template to apply to the project to generate the model;
- creating, based on the model management data and the template, the model, wherein the model comprises a directory, a data structure, and a diagram that describes the project;
- converting the model into a device-agnostic format version of the model; and

- loading, to a common repository, model data that comprises the device-agnostic format version of the model.

15. The computer storage medium of claim 14, wherein the device-agnostic format version of the model comprises hypertext markup language or extensible markup language.

16. The computer storage medium of claim 14, wherein the common repository also stores a model history associ-

ated with the model, wherein the model history comprises data that describes creation of the model.

17. The computer storage medium of claim 14, wherein the computer-executable instructions, when executed by the processor, cause the processor to perform operations further comprising:

- detecting a model update comprising a request to update the model;

- retrieving, from the common repository, the model;

- updating the model based on the model update;

- generating a model update event that defines an update to the model;

- adding the model update event to a model history stored at the common repository; and

- storing an updated model and an updated model history at the common repository.

18. The computer storage medium of claim 17, wherein the model update event comprises an addition to the model, and wherein updating the model comprises:

- obtaining an asset from an asset catalog, wherein the asset relates to the addition; and

- updating the model based on the asset and the model update event.

19. The computer storage medium of claim 14, wherein the computer-executable instructions, when executed by the processor, cause the processor to perform operations further comprising:

- receiving a request for a model history view relating to the model;

- obtaining model data and a model history from the common repository, wherein the model data describes the model and wherein the model history describes lifecycle events associated with the model;

- generating the model history view, wherein the model history view depicts model update events associated with the model and versions of the model; and

- enabling interactions with the model history view by the authenticated user device via portal.

20. The computer storage medium of claim 19, wherein the model history view comprises depictions of at least two model update events associated with the model, the model update events being obtained from the model history, and wherein the model history view provides access to at least two versions of the model.

* * * * *